RP-Sanjiv Goenka Group
Growing Legacies

CESC LIMITED

CM CELL
HANDBOOK

Version 1.0

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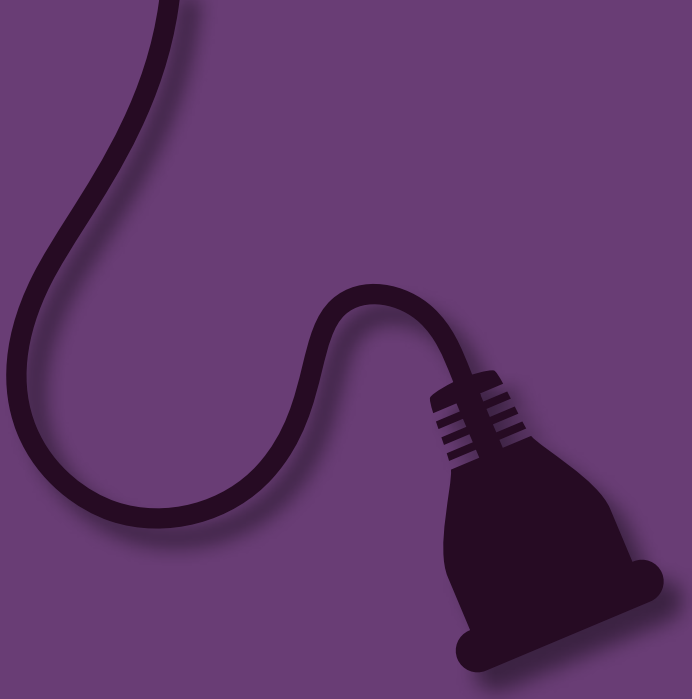
CONDITION MONITORING CELL
2026-2027



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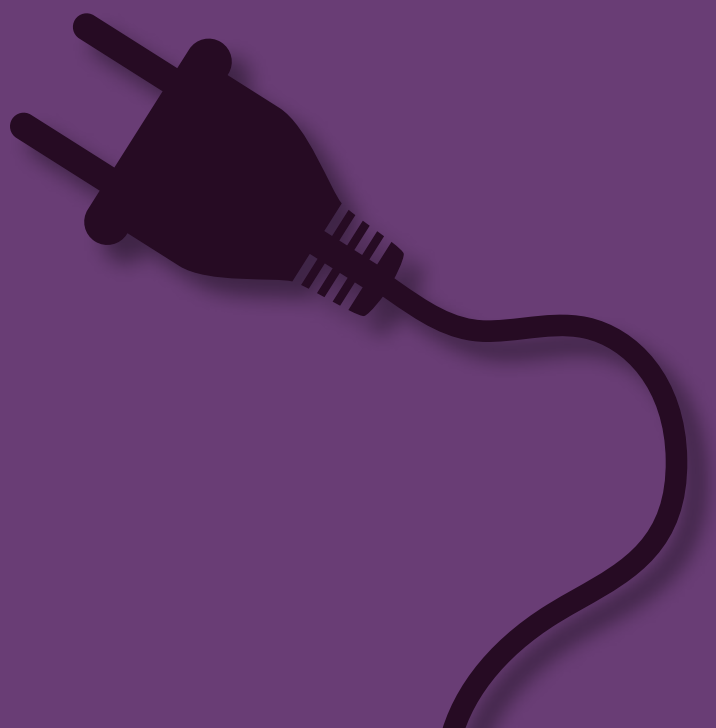

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From the Desk of

GENERAL MANAGER

It gives me immense pleasure to present the Condition Monitoring Cell Handbook, which has been prepared as a comprehensive reference for engineers, supervisors, and technical personnel engaged in the monitoring and maintenance of electrical assets across the CESC Network.

Condition Monitoring has become an integral part of modern power system management. By adopting advanced diagnostic techniques such as infrared thermography, partial discharge detection, earth resistance measurement, and other non-intrusive monitoring methods, the CM Cell aims to detect incipient defects at an early stage and prevent potential failures before they impact system reliability.

This handbook consolidates the standard operating procedures, safety practices, diagnostic methodologies, and inspection workflows followed by the CM Cell. It has been designed not only as a technical guide for field execution but also as a framework to ensure uniformity, discipline, and quality in monitoring activities across both upstream and downstream assets.

I sincerely hope that this handbook will serve as a practical guide for field teams and contribute to strengthening a culture of preventive maintenance, safety, and engineering excellence within our organization.
I appreciate the efforts of all the team members of the CM Cell for their contributions in preparing this document.

Arindam Sanyal
(GM – T&C and CM Cell)
CESC Limited

INTRODUCTION

Condition Monitoring Handbook

PREFACE

1.1 Purpose of the Handbook

The principal objective of this handbook is to articulate a coherent framework of standard operating procedures, best practices, and technical guidance for the systematic condition monitoring of transformers, switchgear, RMUs, cables, capacitor banks, pillar boxes, and other high- and low-voltage installations.

In an era where infrastructure resilience is not merely desirable but indispensable, this compendium seeks to:

- Detect incipient defects at their nascent stage, before they metastasize into disruptive failures
- Facilitate predictive and condition-based maintenance in lieu of reactive intervention
- Institutionalize uniform monitoring practices and disciplined data acquisition
- Minimize forced outages and mitigate safety risks
- Prolong asset longevity while enhancing the overall reliability of the electrical ecosystem

This handbook is therefore not simply procedural in intent; it is philosophical in its advocacy of foresight over firefighting.

1.2 Intended Users

This document is conceived for:

- Field engineers and technicians engaged in on-ground diagnostics
- Supervisors and Condition Monitoring (CM) coordinators entrusted with oversight and quality assurance

It functions simultaneously as:

- An operational compass for field execution
- A technical reference benchmarking for asset health monitoring & evaluation.

It is intended to bridge the often-underestimated gulf between policy and practice, ensuring that theoretical knowledge is translated into disciplined execution.

1.3 Scope of Coverage

This handbook delineates condition monitoring practices across a comprehensive spectrum of electrical assets, spanning both distribution-level infrastructure and high-capacity transmission systems.

1.3.1 Downstream Assets (11 kV/6 kV/3.3 kV/415 V)

This category encompasses distribution network components that form the final conduit of electrical energy to end consumers. Their reliability is paramount, as localized failures here have immediate and tangible repercussions.

Periodic thermographic and diagnostic surveillance within this domain is essential, not merely for fault detection but, for sustaining system stability and consumer confidence.

Assets covered include:

- Distribution Transformers
- Ring Main Units (RMUs)
- Consumer Switchgears
- Consumer Modules
- Packaged Sub-Stations (PSSs)

These installations, though modest in scale relative to upstream systems, are critical nodes in the architecture of dependable supply.

1.3.2 Upstream Assets (High Voltage and Extra High Voltage)

This category addresses the backbone of transmission and high-capacity distribution infrastructure — installations where the consequence of failure is systemic rather than localized.

It includes assets installed at outdoor yards and indoor switchgear of:

- Distribution Stations
- Sub-Stations
- Receiving Stations
- EHV Consumers
- Import Points
- Switching Stations

Within this echelon, reliability is not optional; it is foundational to grid integrity.

The monitoring domains addressed herein encompass:

- **Status Monitoring (SM)** of both upstream and downstream assets
- **Condition Monitoring (CM)** employing advanced diagnostic techniques, including:
  - Infrared thermography
  - Partial discharge monitoring (UHF, RFI, ultrasonic/acoustic methods)
  - Earth Loop resistance measurement
  - Transient Earth Voltage (TEV) detection

The handbook advances a multi-technology, non-intrusive, online monitoring philosophy — one that privileges continuous awareness over episodic reaction. It also provides indicative monitoring periodicities and articulates preparedness protocols for special or high-risk events.

In essence, this document aspires to transform monitoring from a mechanical routine into an intelligent, anticipatory discipline.

> Coming together is a beginning; keeping together is progress; working together is success.
>
> — Henry Ford

Condition Monitoring in CESC Ltd

Background

In 2009, CESC Ltd engaged Singapore Power PowerGrid (SPPG) to conduct a comprehensive study of its distribution system, aimed at modernization and the reduction of unplanned outages. One of the key recommendations of this study was the creation of a dedicated Condition Monitoring Cell (CM Cell), which was subsequently established in July 2010.

Evolution and Role of the CM Cell

Condition monitoring has become a core reliability function within the CESC Ltd distribution network, spanning voltage levels from 220 kV down to 415 V and including a wide base of pillar boxes and consumer installations.

The CM Cell follows a **non-intrusive, no-outage monitoring philosophy**, enabling health assessment of live electrical assets without any interruption to consumer supply.

Failure Mechanisms Addressed

High-voltage and medium-voltage assets are susceptible to failures caused by:

- Overheating and localized hotspots
- Insulation degradation
- Loose, oxidized, or corroded electrical connections
- Partial discharge activity
- Moisture ingress
- Environmental pollution
- Ageing and deterioration of dielectric materials

Early detection of these defects helps prevent:

- Forced outages
- Catastrophic equipment damage
- Fire, arc flash, and personnel safety hazards

Diagnostic Technologies Deployed

The CM Cell deploys a comprehensive suite of advanced diagnostic technologies, including:

Infrared Thermography

- Identification of abnormal temperature rise in transformers, RMUs, switchgear, isolators, conductor joints, bushings, capacitor banks, and LV distribution pillars.

Partial Discharge Diagnostics

- UHF-based PD detection for GIS and metal-clad switchgear
- Ultrasonic/acoustic PD detection for airborne PD and surface discharge
- Transient Earth Voltage (TEV) detection for metal-clad panels, RMUs, and cable terminations
- RFI based PD detection in electrically noisy areas like Outdoor Yards, HV Bushings

Earth Loop Resistance

Measurement of Earth Resistance is to verify the integrity of grounding systems in industrial and utility environments for enhanced safety.

Multi-Technology Monitoring Philosophy

The CM Cell adopts a layered and multi-technology diagnostic approach in which:

- Multiple diagnostic technologies are applied to the same asset
- Results are correlated to improve diagnostic confidence
- Techniques are selected based on asset type, voltage level, accessibility, and criticality

Monitoring activities include:

- Online non-contact measurements
- Use of portable instruments and sensors
- Targeted monitoring during special events and system contingencies

Monitoring Periodicity

- Substations, distribution stations, and EHV consumer installations:
  - Condition Monitoring at **6-month intervals**
- Distribution network assets (**earlier practice**):
  - 11 kV assets – 12 months
  - 6 kV assets – 24 months

This resulted in repeated inspections in the same season.

- Revised monitoring periodicity (**current practice**):
  - 11 kV assets – 10 months
  - 6 kV assets – 22 months

This revision captures seasonal variation effects and improves diagnostic reliability.

Special-event and contingency monitoring is conducted prior to:

- Major festivals, sporting and public events
- Monsoon preparedness
- Sensitive and critical installations during an emergency

SAFETY MANAGEMENT

General Safety Guidelines

(Applicable to all SM & CM activities)

1. Mandatory Personal Protective Equipment (PPE)

The following PPE must be used during all SM and CM activities:

- Safety helmet
- Safety shoes
- Safety gloves (for material handling)
- Safety goggles
- Safety harness for working at heights (above 6 feet)
- Live Line Detector (Downstream)

Personnel must ensure PPE is in good condition before use.

2. Safety Precautions Before Starting the Job

Before commencing any activity, the following precautions shall be ensured:

2.1 Work Planning

- Inform the Power/Distribution Control and appropriate authority prior to commencing the work and obtain all necessary permissions and clearances from the concerned authority.
- Clearly explain the job scope to all team members and ensure that all required instruments and equipment are arranged and available.
- Conduct the Office Toolbox Meeting before proceeding to the site to brief the team on the task and safety requirements.
- Check availability of **First Aid Box** and check its content.
- Check availability and condition of **PPE**.

2.2 Toolbox Meeting (TBM)

A toolbox meeting must be conducted at site before starting work to:

- Explain the job to team members
- Identify hazards and associated risks

- Decide appropriate control measures
  - Elimination
  - Substitution
  - Engineering control
  - Administrative control
  - PPE
- Discuss emergency procedures
- Record unsafe conditions or near-miss incidents if observed

2.3 Site Preparation

- Inform **station in-charge / security personnel** regarding the purpose of visit, work location and expected duration.
- Ensure adequate lighting at the work location.
- Identify **emergency exits and safe escape routes**.
- Display **danger boards and caution boards** to create a safe working zone.
- Ensure **proper earthing connections** where required.
- Verify measuring instruments are properly earthed and supplied through **RCCB protection**.
- Inspect tools and instruments before use.

3. Safety Precautions During the Job

During execution of work, the following safety practices must be strictly followed:

3.1 Personal Safety

- Use PPE during the work and follow the SWP.
- Follow the harness anchoring rules while working at height.
- Maintain statutory safe distance from live electrical parts.
- Do not stand on live electrical equipment.
- Do not insert measuring probes or instruments inside live chambers.
- Before opening, check the hinge condition of LT doors of DTR, FPB and RMU.
- While opening LT doors, do not stand directly in front of the door.
- Do not use mobile phone unnecessarily at site.
- Always ensure one team member monitors the surroundings while the other performs the task

3.2 Electrical Safety

- Visually check the earthing of the equipment under inspection.
- Use line tester before starting the inspection.
- Do not touch **GIS PD sensors during switching operations**.
- Ensure isolated systems are **properly earthed during PD resolving activities**.
- Maintain safe distance during **HV injection testing and outdoor yard**.
- CT secondary circuits must be **short-circuited and properly earthed before testing**.
- Do not step on **HT cable joints, AIS circuit and busbar chambers**.

3.3 Environmental Safety

- Avoid working in **outdoor yard during stormy weather**.
- Avoid working in water logged condition.
- Remain alert while working near **open cable ducts, trenches and working at cable basement.**
- Avoid contact with **hanging kite strings or foreign objects near electrical equipment**.
- Take measures to alert surrounding people while working in public area.
- If live part of any equipment is found exposed or against public safety then inform immediately. That equipment may not be in your inspection list.

3.4 Equipment Safety

- Ensure measuring instruments are correctly connected and handled.
- Do not operate equipment showing visible defects.
- Use proper earthing, safe power connection and RCCB protection while charging or inspection.
- Do not try to repair any defective equipment.

4. Safety Precautions After Completion of Job

After completing the activity:

- Inform **Station In-Charge / Control Room** about completion of work.
- Ensure all equipment panels, doors and covers are properly closed.
- Remove tools, instruments and safety boards from the work area.
- Restore the site to a **safe and normal operating condition**.
- Report any **abnormality, unsafe condition and incident** observed during the work.

5. Equipment Isolation and Grounding Rules

- Routine **SM and CM activities generally do not require equipment isolation**.
- During **PD resolving**, voltage shall be applied **phase-wise while keeping other phases earthed**.
- During **PD testing of isolated CTs**, CT secondary circuits must be **short-circuited and properly earthed**.

6. Statutory Safety Distance

VOLTAGE LEVEL	GROUND CLEARANCE	HORIZONTAL CLEARANCE
11 kV	2.75 m	2.60 m
33 kV	3.70 m	2.80 m
132 kV	4.60 m	3.50 m
220 kV	8.00 m	6.50 m

Safe working zones shall be established using **caution boards and barricading**.

7. Emergency Procedures

In case of emergency:

- Immediately report unsafe conditions or accidents.
- Provide **first aid and arrange medical assistance** if required and register it.
- Keep 2nd Safety harness for rescue operation.
- Do not use lifts during **evacuation situations.**

> Efficiency is doing things right; effectiveness is doing the right things.
>
> — Peter Drucker

Job-specific Safe Working Procedures

1. Cable Basement – Upstream SM & CM Activity

Before Entering Cable Basement

- Inform station in-charge / security personnel about the work.
- Obtain System Control permission.
- Conduct TBM and identify unsafe conditions.
- Check PPE, First Aid Box, ID card, Authorisation Letter.
- Ensure **CO_2 system is switched OFF**.
- Ensure **exhaust fan or blower is running & w**ait at least **5 minutes before entering confined space**.
- Check for abnormal smell in the basement.
- Ensure adequate lighting inside the basement.
- Check for **water logging** and drain out if present.
- Use caution board for safe zone creation.

During Work

- Use appropriate PPE.
- One team member must remain on watch while the other performs the activity.
- Identify and avoid stepping on **existing cable joints**.
- Remain alert to **cable trays, ducts and structural obstructions**.

After Completion

- Inform Station In-Charge / Control Room.
- Notify all unsafe conditions and incidents.

2. Indoor Switchboard – Upstream SM & CM Activity

Before Starting Work

- Inform Station In-Charge / Control Room.
- Conduct TBM and identify unsafe conditions.
- Check PPE, First Aid Box, ID card, Authorisation Letter.
- Use caution board for safe zone creation.
- Ensure adequate illumination inside the switch room.
- Maintain safe distance from **exposed HT chambers**.

During Work

- Use appropriate PPE.
- Use line tester.
- Visually check earthing system of the Switchboard.
- Do not stand on Circuit and Bus Bar chamber.
- Do not open any HT chamber.
- Do not perform any unauthorised operation or activity.
- Do not insert sensor probe inside HT chamber.
- Maintain safe distance while taking measurements.
- Stay alert to **open cable ducts.**

After Completion

- Inform Station In-Charge / Control Room.
- Notify all unsafe conditions and incidents.

3. Outdoor Yard (ODY) – Upstream SM & CM Activity

Before Starting Work

- Inform station in-charge / security personnel.
- Inform Control Room before commencing activity
- Conduct TBM and identify unsafe conditions.
- Check PPE, First Aid Box, ID card, Authorisation Letter
- Use caution board for safe zone creation.

During Work

- Use appropriate PPE.
- Use line tester.
- Keep a safe distance from live parts.
- Do not touch any object or kite wire hanging from live parts.
- Do not stay inside the ODY during stormy or rainy weather.
- Do not use umbrella inside ODY.
- Remain cautious of **open cable ducts and trenches.**
- One team member must remain alert while the other performs work.

After Completion

- Inform Station In-Charge / Control Room.
- Notify all unsafe conditions and incidents.

4. RMU / Switchgear – Downstream SM & CM Activity

Before Starting Work

- Conduct TBM and identify unsafe conditions.
- Check PPE, First Aid Box, ID card, Authorisation Letter.
- Use caution board for safe zone creation.
- Ensure personnel carry **standard RMU keys**.
- Inform System Control before working on **automated RMUs**.

During Work

- Use appropriate PPE.
- Use line tester.
- Keep a safe distance from live parts.
- Check hinges and then open RMU doors for inspection.
- Remain cautious of open cable ducts and trenches.

After Completion

- Inform Station In-Charge / Control Room.
- Notify all unsafe conditions and incidents.

5. OT / TH / PT Structure – Downstream SM & CM Activity

Before Starting Work

- Conduct TBM and identify unsafe conditions.
- Check PPE, First Aid Box, ID card, Authorisation Letter.
- Use caution board for safe zone creation.
- Inspect ladder and scaffolding of P/T structure before climbing.
- Verify sign writing on **transformer body, HT box and LT box**.
- Ensure both personnel carry **standard keys**.

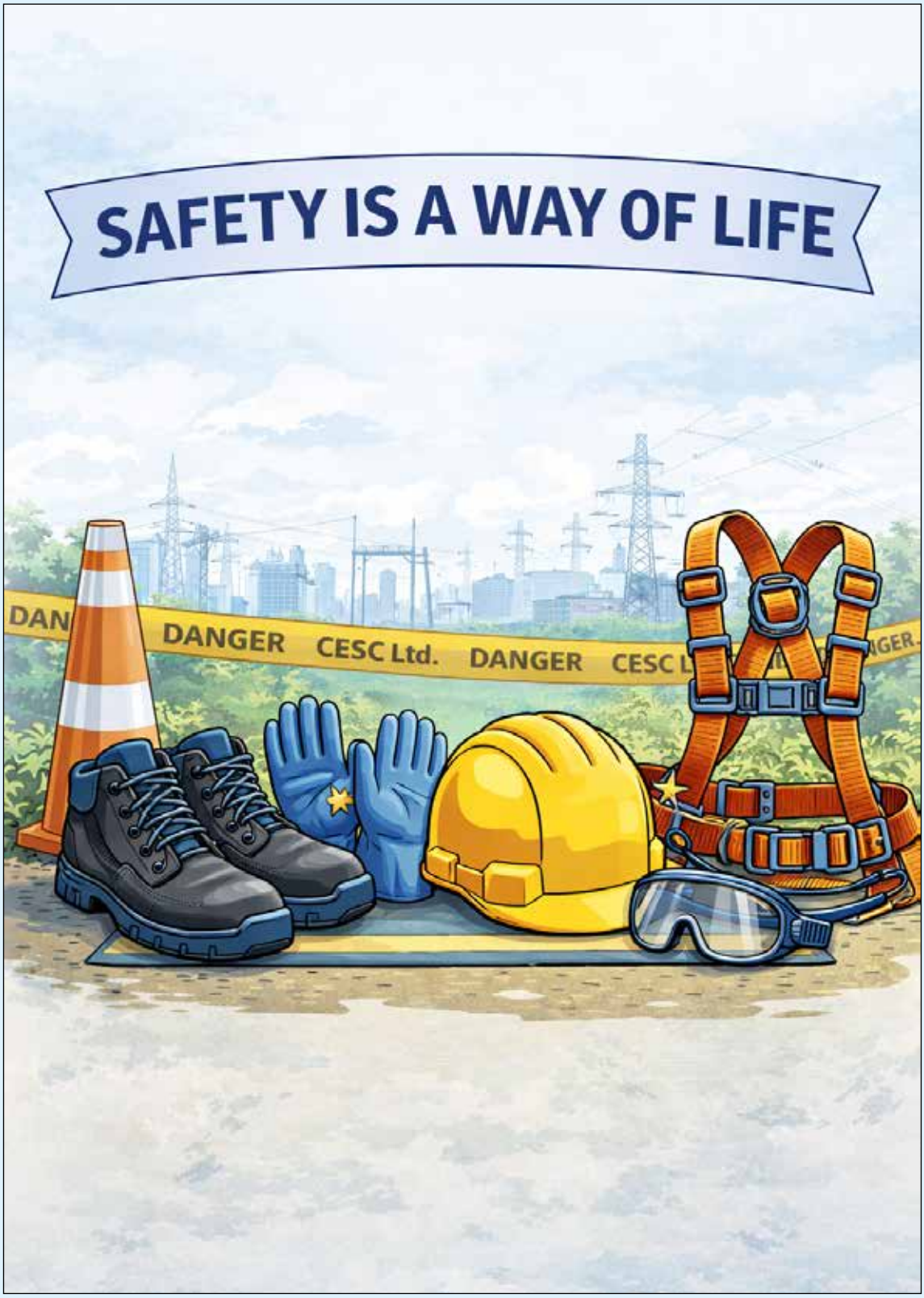
During Work

- Use appropriate PPE.
- Use line tester.
- Keep a safe distance from live parts.
- Visually check the hinges and then open DTR LT doors for inspection.
- Do not stand in front of the LT door or FPB door while opening it.
- Remain cautious about open type live FPB inside T/H room.
- While climbing the P/T through CAT ladder follow the safety rule of anchoring the safety harness.
- Remain cautious about vehicles while inspecting road side assets.

After Completion

- Notify all unsafe conditions and incidents

> Success is the result of perfection, hard work, learning from failure, loyalty, and persistence.
>
> — Colin Powell


SAFETY IS A WAY OF LIFE
DANGER CESC Ltd. DANGER

ASSET CONDITION MONITORING

Partial Discharge (PD) Monitoring Technology

As per IEC 60270:2000, partial discharge is a localized electrical discharge that partially bridges the insulation between conductors.

Although PD pulses are very short in duration (typically less than 1 μs), their cumulative effect causes:

- Chemical degradation
- Thermal damage
- Mechanical erosion

Sustained PD activity significantly reduces insulation life and may ultimately result in final insulation breakdown, arc flash, fire, or equipment explosion.

4.2 Types of Partial Discharge (By Location)

- **Internal Partial Discharge** – Occurs within cavities or voids in solid insulation
- **Surface Partial Discharge** – Occurs along insulation surfaces and interfaces

4.3 Causes of Partial Discharge

- Surface contamination and pollution
- Poor workmanship and installation defects
- Manufacturing defects
- Improper component selection
- Mechanical damage during installation or service
- Ageing and deterioration of insulation

4.4 Classification of Partial Discharge (By Nature)

- **Corona** – Ionization of air or fluid surrounding a conductor
- **Tracking** – Surface discharge over contaminated insulation
- **Arcing** – Complete electrical breakdown producing plasma

Physical indicators include:

- Ozone or burning smell
- Discoloration of insulation
- Carbon tracking marks
- Presence of 40 kHz ultrasound

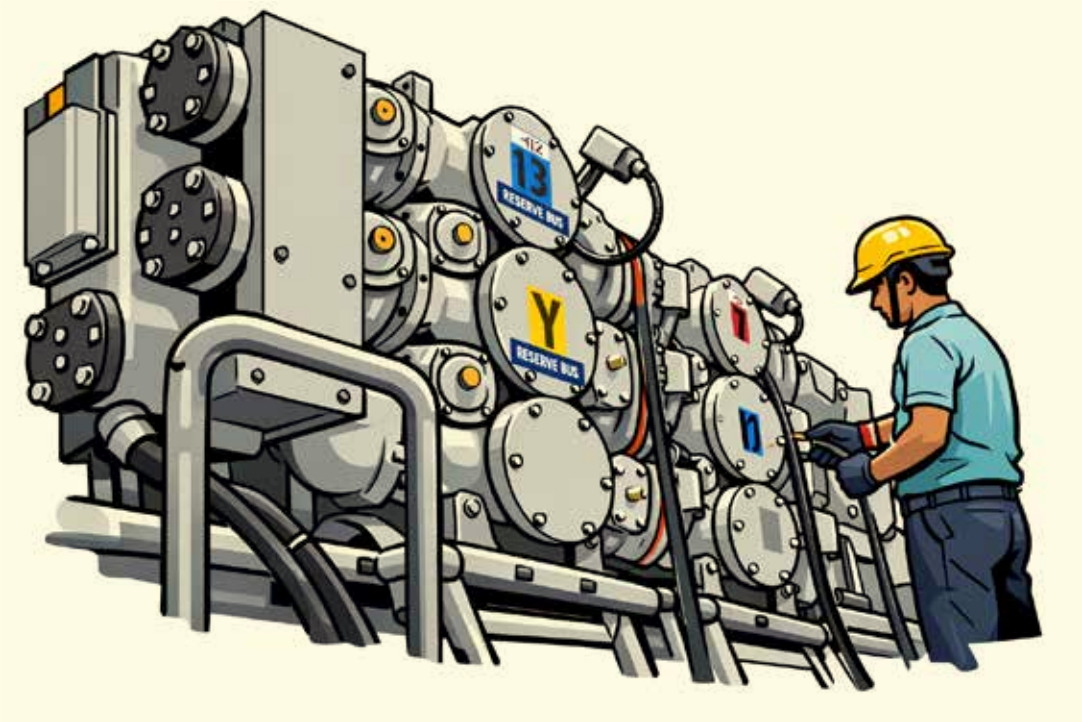
4.5 Non-Intrusive PD Monitoring (No-Outage Testing)

Non-intrusive PD monitoring enables online testing without de-energizing equipment.

The techniques include:

- **Ultrasonic / Acoustic** testing - Through louvers, vents, inspection windows, air gaps between joints & hinges
- **TEV testing** - Detection of surface PD in metal-clad panels and RMUs
- **RFI detection** -Directional and non-directional RF sensors to detect radiated PD energy
- **UHF** (Ultra High Frequency)**:** Detection of internal high frequency in the range of UHF zone up to 1500 MHz frequency signals.

These methods collectively form the core of **no-outage, online PD Monitoring**.



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DIAGNOSTIC INSTRUMENTS FOR CONDITION MONITORING

PD Monitoring Measuring Instruments

A. Airborne Ultrasonic Discharge Activity

Acoustic emissions produced by **partial discharge (PD)** occur over a wide range of the acoustic spectrum. Although some components fall within the audible range, detection by human hearing depends on individual sensitivity and surrounding noise levels.

Discharge phenomena such as **arcing, tracking, and corona** involve ionization processes that disturb nearby air molecules. This molecular agitation generates predominantly **high-frequency acoustic signals**, often extending into the **ultrasonic range (>20 kHz)**.

Ultrasonic monitoring instruments use **piezoelectric sensors** to detect airborne emissions above the audible spectrum. These sensors are highly directional and effective at capturing the high-frequency sounds produced by ionization. The **intensity of the detected sound** depends on the type of discharge and the acoustic impedance of the medium. The most sensitive detection method uses an **airborne ultrasonic microphone**, typically applied at accessible locations such as **insulators, bushings, and cable terminations**.

Ultrasonic PD measurement relies on both **qualitative audio analysis** and **signal intensity levels** indicated by the instrument. Studies show that airborne ultrasonic instruments are particularly effective for detecting **corona, tracking, and arcing**, although detection may require sustained emission for the microphone to capture the signal. Ultrasonic sensors can also transmit through the **solid structure of equipment components**.

Key characteristics of ultrasonic discharge detection:

- Influenced by environmental factors (temperature, humidity, pollution)
- Partial discharge typically produces a **distinct intermittent crackling sound**

Ultrasonic Discharge Measuring Instruments

Purpose

- Specifically designed to detect partial discharge activity in HV and EHV equipment.
- Facilitates predictive maintenance by identifying early-stage component degradation before it results in unplanned outages.
- Also used for leak detection (compressed air, steam, and vacuum systems) and mechanical inspections (e.g., bearings and valves) to support improved energy efficiency and asset reliability.

UE Systems Make - ULTRAPROBE 9000

(Digital Ultrasonic Inspection System)

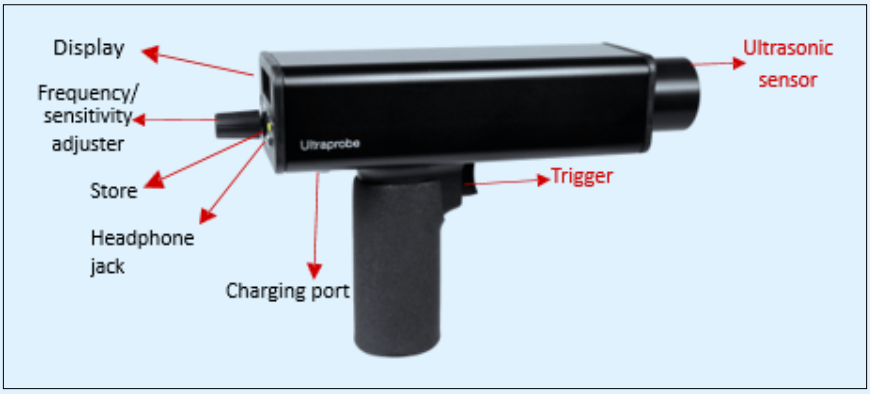


Fig 1: *UP 9000 with short range ultrasonic sensor*

Fig 2: *Different components of UP 9000*

Device Interface

- **Trigger Switch:** Pull and hold to power the unit and begin active scanning. Releasing the trigger enters "Standby" mode, which freezes the current decibel reading and holds it on the display.
- **Spin and Click Control:** A multifunctional rotary dial used to adjust sensitivity levels in real-time. When clicked, it allows the user to enter the menu to adjust frequency (20–100 kHz), store data, or change display modes.
- **Backlit LCD Display:** A digital screen providing a calibrated decibel (dB) readout, frequency, and battery life indicator. It also displays stored record numbers.
- **Plug-In Modules:** Features a quick-connect port for interchangeable sensors, including the Tri-sonic Scanning Module (airborne), the Stethoscope Module (contact-borne) and magnetic mountable sensors.
- **Headphone Jack:** Standard output for high-quality, noise-attenuating headphones, allowing the user to listen to the heterodyned ultrasonic signal for diagnostic audio.
- **I/O Port (Dual Function):** A high-speed interface used for recharging the internal battery and connecting to a PC via USB for downloading saved inspection data to the Ultratrend DMS software.

Principle of Operation

The UP9000 uses a piezoelectric transducer with interchangeable modules to detect high-frequency sound waves (ultrasound) between 20 kHz and 100 kHz. It converts these inaudible ultrasounds to a lower, audible frequency (heterodyned output) for the user to hear via a noise-attenuating headset. Concurrently, it displays the sound intensity on a calibrated decibel (dB) readout and a 16-segment bar graph for data logging and trending.

Key Specifications

PARAMETER	SPECIFICATION
Type of Inspection	Digital Ultrasonic (Airborne and Contact)
Frequency Range	20 kHz to 100 kHz (Tuneable in 1 kHz increments)
Decibel Range Frequency Range	120 dB Dy namic Range (Calibrated dB readout)
Memory/Storage	400 storage locations
Data Output	Decibel (dB), frequency, USB data output
Power	NiMH Rechargeable Battery

Operation Steps (SOP)

1. Make sure the battery is charged.
2. Insert the correct module (LRM, Short range sensor, Magnetic mountable sensor etc.)
3. Insert the headphone jack.
4. Press and hold the on button to start and set the frequency to 20-25 kHz for centaca mode and 40 kHz for airborn pastial discharge.

Accessories

1. **Standard Detection Range (Tri Sonic Module):** The Ultraprobe 9000's standard module can detect from a distance of 15.24 m.
2. **Long Range Module:** The LRM-9 doubles this distance for airborne emissions, its practical range for this specific leak sensitivity benchmark is extended to approximately 30.48 m. The instrument should be set to 40 kHz frequency for detecting airborne partial discharge activity while using Tri-sonic and LRM modules.
3. **Magnetic Mountable & Contact Mode Sensors:** For condition monitoring, the sensor must be in direct physical contact with the machine's surface (e.g., CT, VT secondary housing) to pick up the high-frequency vibrations (ultrasound) traveling through the solid structure. The instrument should be set within the 20–25 kHz range to capture the mechanical sound generated by electrical discharge activity inside the equipment tank. The magnetic base provides a fast, secure, and temporary means of establishing consistent mechanical coupling with the equipment surface, ensuring reliable signal transmission during measurement.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power	Check battery charge or pull/release the trigger to reset the power.
Inaccurate dB Readings	Adjust the frequency tuning (20–100 kHz) to filter out background plant noise, or check that the correct module (Airborne/Contact) is used.
High Ambient Noise Interference	Use the Rubber Focusing Probe to narrow the reception field, or increase the frequency setting to isolate the target ultrasound.
No sound in headset	Verify the Headset is plugged securely into the jack and the volume/ sensitivity dial is adjusted correctly.
Data Not Stored	Check the unit's 400 location memory status. If full, download and clear data via USB to the Ultratrend DMS software.

Basic Troubleshooting

Frequency Selection	Spin and Click Control > Spin Dial to set Frequency 25 or 40 kHz as per requirement.
Data Storage	Press Yellow Store button to log the current Db, Frequency, Time, Date & text data to memory.
Display Mode	Press the trigger>Point at the particular location> Release the trigger> Capture Real time data hold.

UE Systems Make - ULTRAPROBE 3000

(Digital Ultrasonic Inspection System)

Device Interface

- **Trigger Switch**: Pull and hold to power the unit; release to enter "standby" mode and hold the current reading.
- **Spin and Click Control:** A rotary dial used to adjust sensitivity levels and enter/select menu options.
- **LCD Display**: Backlit digital screen showing decibel (dB) levels, sensitivity, and battery status.
- **Plug-In Modules:** Interchangeable sensors for airborne (Scanning) or structure-borne (Stethoscope) detection.
- **Headphone Jack:** Output for high-quality noise-attenuating headphones to hear heterodyned sound.
- **I/O Port:** Dual-purpose port for charging the battery and connecting the USB cable for data transfer.

Principle of Operation

- The Ultraprobe 3000 utilizes a piezoelectric transducer to sense ultrasonic sound waves in the 35 kHz to 45 kHz range.
- The system uses electronic circuitry to down-convert these high-frequency signals into the audible range through a process called heterodyning.
- The converted audio is delivered via headphones, while the intensity of the signal is converted into a digital decibel (dB) value displayed on the LCD for objective measurement.

Key Specifications

PARAMETER	SPECIFICATION
Type of Inspection	Digital Ultrasonic (Airborne and Contact)
Frequency Range	Fixed at 35 kHz to 45 kHz (centred at 40 kHz)
Decibel Range	0 to 70 dB
Memory/Storage	400 storage locations
Data Output	Decibel (dB), USB data output
Power	NiMH Rechargeable Battery

Operation Steps (SOP)

1. **Check Battery:** Pull the trigger and check the battery level icon on the LCD display.
2. **Select Module:** Insert the Scanning Module for air/gas leaks or the Stethoscope (Contact) Module for internal mechanical testing.
3. **Connect Audio:** Securely plug the noise-attenuating headphones into the jack at the rear of the pistol.

4. **Scanning:** Pull the trigger and point the sensor toward the test area. Adjust the Spin Dial to change the sensitivity if the decibel reading is too high (at maximum) or too low.
5. **Record Data:** Once a reading is stabilized, click the Spin Dial to save the decibel level to a specific memory location.

Accessories

1. **Scanning Module:** Standard sensor for detecting airborne ultrasound such as pressure leaks or electrical discharge.
2. **Stethoscope Module:** A contact-type probe used to listen to internal sounds like bearing friction or steam trap cycles.
3. **Rubber Focusing Probe:** A cone-shaped attachment that slides over the scanning module to shield against ambient noise and narrow the detection field.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power	Ensure the battery is charged. If it still won't turn on, try resetting by removing the battery for 30 seconds.
Fixed "0" dB Reading	Sensitivity is likely set too low. Spin the dial clockwise to increase the sensitivity level.
High Background Noise	Switch to the Rubber Focusing Probe or use the "shielding" technique (using your body to block competing sound sources).
Poor Audio Quality	Check the headphone connection. Ensure the sensor module is seated in its position.
Memory Full	The 400-location memory must be cleared. Connect to a PC and use the DMS software to download and wipe the memory.

Basic Troubleshooting

FEATURE	ACTION
Sensitivity Adjustment	Spin the Rotary Dial up or down to adjust the sensitivity (displayed as 0–70 on the LCD).
Data Storage	Click the Spin Dial (press inward) to save the current dB reading to the selected memory location.
Memory Navigation	Pull the trigger to enter the menu, then spin the dial to scroll through stored data locations (1–400).
Data Transfer	Connect to a PC via the USB/Charging port and select the "Transfer" option in the DMS software.

UE Systems Make - ULTRAPROBE 100

(Analogue Ultrasonic Inspection System)

Device Interface

- **Trigger Switch:** Pull and hold to power the unit; release to turn off (does not have a "standby" hold feature).
- **Max dB dial:** A 0 to 70 dB turn notched potentiometer dial used to adjust max amplitude of the ultrasonic signal.
- **LED Bar Graph:** A 10 segment LED display that indicates the intensity of the ultrasonic signal (Red LEDs).
- **Battery Light:** A dedicated LED that illuminates when the battery is low or requires replacement.
- **Plug-In Modules:** Interchangeable sensors for scanning (airborne) or stethoscope (contact) applications.
- **Headphone Jack:** Output for high-quality noise-attenuating headphones.

Principle of Operation

- The Ultraprobe 100 utilizes a piezoelectric transducer to detect high-frequency sound waves in the ultrasonic range.
- The instrument has a peak response frequency within 36-44 kHz, which is the optimal frequency for most partial discharge detection.
- It employs heterodyning technology to convert these inaudible sounds into an audible signal heard through headphones, while simultaneously showing the intensity on an analogue-style LED bar graph.

Key Specifications

PARAMETER	SPECIFICATION
Type of Inspection	Analog Ultrasonic (Airborne and Contact)
Frequency Range	36-44 kHz
Display	10-Segment LED Bar Graph
Max dB control	0 to 70 dB-turn notched dial
Memory/Storage	None (Real-time monitoring only)
Data Output	Heterodyned audible signal
Power	9 V Alkaline Battery

Operation Steps (SOP)

1. **Insert Battery:** Ensure a fresh 9 V alkaline battery is installed in the handle.
2. **Attach Module:** Plug in either the Scanning Module (for leaks/electrical) or the Stethoscope Module (for internal mechanical).
3. **Connect Headphones:** Plug the headset into the jack before powering the unit.
4. **Power On:** Pull the trigger switch. The LED bar graph should flicker briefly.
5. **Adjust Sensitivity:** Turn the sensitivity dial until the LED bar graph is at approximately mid-range while scanning the target area.
6. **Locate Source:** Follow the sound and the increasing LED display to the point of highest intensity (the "loudest" point).

Accessories

1. **Rubber Focusing Probe:** A rubber shield narrows the scanning module's field of view and block ambient ultrasound.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power / No LEDs	Check that the battery terminals are making contact else replace the 9 V alkaline battery.
LEDs Stay at Maximum	Sensitivity dial is set too low; turn the dial counter-clockwise to increase gain.
High White Noise	Ensure the sensor module is pushed firmly into the pistol housing.
Low audio	Check if the 9 V battery is weak; the "Low Bat" light may not always trigger if the battery power is in borderline

FLIR Si124 Pro

(Acoustic Imaging Camera)

The FLIR Si124 Pro (and its specialized versions like the Si124-LD for leaks and Si124-PD for partial discharge) is a high-performance acoustic imaging system designed to "see" sound. It uses a massive array of microphones to pinpoint issues up to 10 times faster than traditional ultrasonic scanning methods.

Device Interface

- **Acoustic Sensor Array:** Features 124 low-noise MEMS microphones that detect and triangulate ultrasonic sound sources.
- **Display:** 5-inch high-brightness (1000 cd/m²) resistive touchscreen for easy viewing in outdoor or bright conditions.
- **One-Handed Operation:** Ergonomically designed lightweight pistol grip (approx. 1.08 kg) for safe, easy use on-site.
- **Connectivity:** Built-in Wi-Fi for automatic cloud uploads and a USB port for manual data transfer.

Purpose

- **Partial Discharge (PD) Monitoring:** Identifies high-voltage electrical faults like corona, tracking, and arcing from a safe distance.
- **Predictive Maintenance:** Helps avoid unplanned downtime by detecting faults that are inaudible to human hearing.
- **Compressed Air Leak Detection:** Locates and quantifies pressurized leaks in industrial air systems to reduce energy waste.

Principle of Operation

- The camera uses its 124 microphones to capture sound waves and creates a “sound map” (acoustic image).
- This sound map is overlaid in real-time onto a standard digital camera image, allowing the user to see exactly where the sound is coming from.
- The device uses AI-based analytics to filter out common industrial background noise, focusing specifically on the frequency signatures of leaks or electrical discharge.

Key Specifications

PARAMETER	SPECIFICATION
Microphone Array	124 MEMS microphones, real-time visualization
Bandwidth	2 kHz to 65 kHz (adjustable range)
Distance Range	0.3 m (1 ft) up to 130 m (430 ft)
Dynamic Range	>120 dB
Leak Detection	Real-time quantification in l/min or CFM
PD Classification	Corona, Floating, Surface, or Internal discharge
Memory	32 GB Internal (approx. 2000 snapshots)
Battery Life	Up to 7 hours (external) or 2.5 hours (internal/cordless)

Operation Steps (SOP)

1. **Power On:** Press the power button and wait for the system to boot to the live imaging screen.
2. **Mode Selection:** Choose either Leak or PD mode depending on your inspection task.
3. **Adjust Frequency:** Use the touchscreen to set the filter (typically 20–30 kHz for air leaks) to eliminate background noise.
4. **Scan Area:** Point the camera at components. Look for the “heat map” of sound on the screen.
5. **Capture and Log:** Pull the trigger to save a snapshot. The camera will automatically estimate leak size or PD severity.
6. **Report:** Sync with FLIR Acoustic Camera Viewer (Cloud) to generate professional reports.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
Inaccurate Leak Rates	Ensure the Auto Distance tool is active or manually enter the correct distance to the target.
Image Not Syncing	Check Wi-Fi connection strength or use the included USB stick for manual export.
Battery Draining Fast	Reduce screen brightness or ensure the device is not left in active scanning mode when not in use.

Qualitrol make Portable PDM

(Partial Discharge Monitor for Transformers and GIS)

Fig-1: *Channel 1-6 UHF Signal Inputs, Noise Input, AC Supply, Ethernet Port, Sync Input*

Fig-2: *UHF receiving sensor*

Fig-3: *Noise generator*

Purpose

This instrument is intended for the accurate assessment to analyse the partial discharge (PD) in gas-insulated switchgear (GIS) and air-insulated switchgears (AIS). It supports both online and offline periodic testing, commissioning verification and assist in decision making for condition-based maintenance (CBM), without the need for permanent installation.

Principle of Operation

The Qualitrol Portable PDM operates on **Ultra-High Frequency (UHF)** technology, measuring signals across a wideband range of **300 – 1500 MHz** It detects electromagnetic waves generated by partial discharge events using up to 6 UHF sensors. The system utilizes multiple **Artificial Neural Networks (ANN)** to classify events and differentiate PD from noise signals. It compares detected signals against a built-in reference library of historic PD data to characterize faults (e.g., voids, particles) and allows for approximate localization through amplitude comparison.

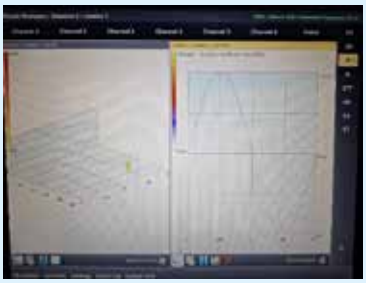
Key Specifications

PARAMETER	SPECIFICATION
Input Channels	6 UHF Channels + 1 Noise Gating Channel
Bandwidth	Wideband 300 – 1500 MHz
Sensitivity	–75 dBm / < 5 pC
Sample Rate	15.4 KS/s per channel (Total 107.5 KS/s)
Display	12-inch LCD Touch Screen (1024 x 768)
Data Storage	16 GB Internal (Expandable to 32 GB)
Power Supply	90–264 VAC (47-63 Hz) or 40–290 VDC
Operating Temperature	–25 °C to +55 °C
Enclosure Rating	IP66 (Rugged, NEMA rated)
Weight	Approx. 17 kg (37.5 lbs)

Operation Steps (SOP)

1. **Deploy:** Roll the unit to the site using trolley wheels and open the IP66 protective case.
2. **Connect:** Ensure the instrument is securely connected to a verified **earth ground** and operate the test button in RCCB to ensure the device is functioning correctly prior to engaging the power supply to prevent electrical hazards.
3. **Power On:** Switch on the unit. The pre-installed software will load automatically (no separate laptop required.
4. **Configure:** Use the touch screen to select **substations1, location 1 and then active channels** and apply **3d and phase resolved** view.

5. **Connect Sensor:** Connect one end of the signal cord (coaxial cable) to the corresponding active channel input on the back side of the instrument. Connect the other end of the cord to the UHF sensor.
6. **Functionality Verification:** Select the target active channel. Apply a signal using an external pulse generator to simulate noise/PD. Verify the signal appears on the screen to ensure the device is running fully functionally and correctly.
7. **Monitor:** Monitor real-time **PRPD (Phase Resolved Partial Discharge)** and **3D patterns** to characterize defects by positioning the UHF sensor at the spacers on circuit breaker chamber, circuit side chamber, and busbar compartment. To mitigate significant background interference, **500 MHz, 900 MHz high-pass and 880-960 MHz band stop filters** are utilized to isolate PD-specific signals. This filtration suppresses low-frequency noise, enabling the instrument to accurately identify and characterize the frequency of discharge events.
8. **GIS Commissioning:** During GIS commissioning, the synchronizing signal derived from the connected HV source must be connected to the instrument's synchronization channel to ensure accurate generation of the PRPD pattern.
9. **Data Export:** Insert USB Drive > Select **Event List** > Select Data > Press **Export**



Common Errors / Troubleshooting Tips

ISSUE	REMEDY
System Fault Alarm	Check the "Self-Check" log for internal hardware faults; verify power source stability.
High Interference/ Noise	Enable "Noise Gating" in the software; ensure the external noise antenna is positioned correctly.
No PD Signal Detected	Verify sensor connections; Check "Detect Threshold" settings; Ensure sensors are mounted internally or on valid inspection hatches.
Touch Screen Unresponsive	Restart the unit; Check for debris on the screen surface or operate with mouse.
Channel Fault	Swap sensor cables to verify if the fault is with the cable/sensor or the input port.

B. TEV (Transient Earth Voltage)

Transient Earth Voltages (TEVs) are short-duration, high-frequency voltage spikes that appear on earthed metal surfaces due to internal partial discharge (PD) in high-voltage equipment.

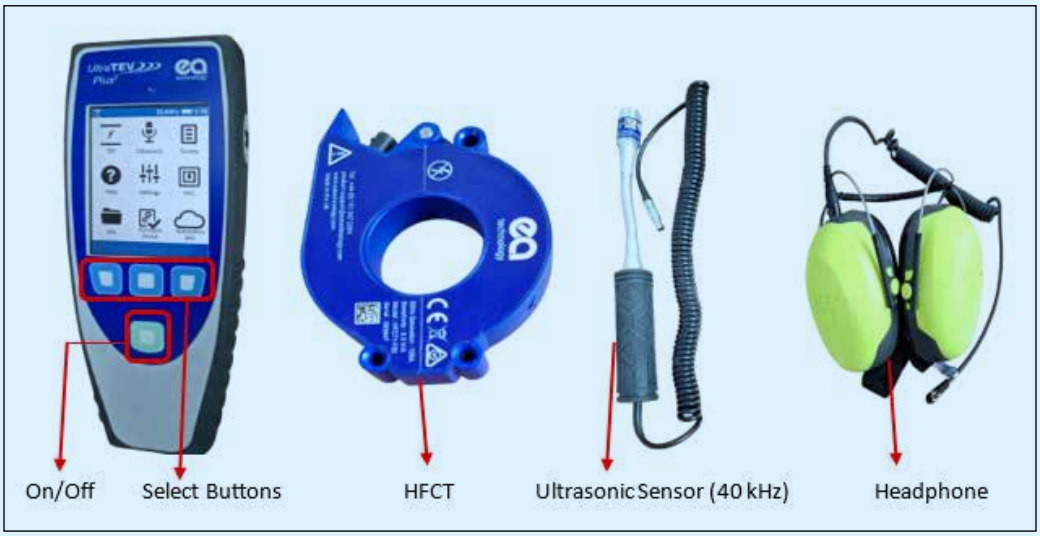
When PD occurs in phase-to-earth insulation, a small charge is rapidly transferred to the surrounding metal enclosure, generating electromagnetic waves and high-frequency current pulses.

Because of the skin effect, these transients travel along the inner metal surface and emerge at discontinuities such as joints or gaskets, creating measurable voltage spikes. TEVs—typically a few millivolts—are measured in dB using a non-intrusive probe on the external switchgear surface while the equipment remains in service.

Tev Measuring Instrument

EA Technology UltraTEV Plus²

(Handheld Partial Discharge Detector)



Controls & Interface

- **Touchscreen:** Menu-driven backlit colour display for navigation and data visualization.
- **Keypad:** Physical buttons designed for easy use while wearing gloves.
- **Power On/Off:** Button to activate/deactivate the instrument.
- **Headphone Jack:** 3.5 mm stereo socket for audible discharge monitoring.
- **Micro USB / SD Slot:** Ports for data transfer and charging.

Purpose

- Used for non-intrusive detection, measurement, and analysis of Partial Discharge (PD) in high-voltage (HV) assets like switchgear, cables, and overhead lines.
- Enables early identification of insulation defects (internal, surface, or corona) to prevent catastrophic failures and optimize maintenance cycles.

Principle of Operation

- The UltraTEV Plus² utilizes multiple sensing technologies: TEV (Transient Earth Voltage) sensors detect internal electromagnetic discharge, while Ultrasonic sensors detect surface tracking and arcing through airborne sound waves.
- Advanced algorithms differentiate between genuine PD activity and background electrical noise, providing clear classification and interpretation of the asset's condition.

Key Specifications

PARAMETER	SPECIFICATION
TEV Measurement Range	0 – 60 dBmV
Ultrasonic Range	-7 dBµVto 71 dBµV
Cable PD (HFCT) Range	100 – 100,000 pC
Display	Backlit colour touchscreen
Battery	Rechargeable Li-ion, ~8 hours continuous use
Storage	8 GB MicroSD Card
Connectivity	Wi-Fi, USB, NFC
OperatingTemp	-20 °C to +50 °C
IP Rating	IP42 (standard) / IP54 (certain configurations)

Operation Steps (SOP)

1. **Power On:** Press the power button; the device features a quick startup for immediate action.
2. **Select Mode:** Choose Survey Mode for guided inspections or Measure Mode for specific TEV/Ultrasonic analysis.
3. **Capture Data:** Press the record button to capture phase plots, waveforms, and histograms of PD activity.
4. **TEV Mode:** Select TEV>Phase Plot>Menu>Select TEV
5. **Ultrasonic:** Connect Headphones >Ultrasonic
6. **Analyse:** Use on-screen tools to differentiate between noise and real PD in real-time.
7. **Save & Transfer:** Save results to the SD card and transfer data to a PC via Wi-Fi or USB for expert review.

Extension Modules & Accessories

- **Ultrasonic Contact Probe:** For detecting surface PD in sealed chambers.
- **Flexible Sensor (Flexi-Probe):** Reaches PD in tight or restricted spaces.
- **HFCT:** Clamps around cable earth grounds to measure PD in cables.
- **Humidity Sensor:** For measuring ambient humidity.

Common Errors / Troubleshooting

ISSUE	REMEDY
No Power	Check battery level; ensure the charging LED is active when plugged in.
Excessive Noise	Use built-in filters or rotate the UHF antenna to isolate interference.
Port Not Recognized	Ensure accessories are firmly clicked into the Lemo ports.
Sync Failed	Verify Wi-Fi connection or check if the MicroSD card is full/unlocked.

HFCT for Detection of Cable PD

HFCT sensors (also known as RFCTs) are widely used for effective detection, identification, and localization of partial discharge (PD) sources.

They operate as induction coils with an electromagnetic core, designed to measure high-frequency transient currents associated with PD pulses. For online PD monitoring of HV equipment, HFCT sensors are clamped around grounded earth-screen conductors while the system remains in service.

As PD pulses propagate through the grounding path, the HFCT detects the resulting high-frequency current signals. The measurement system typically incorporates a tuned circuit with a standard 50-ohm input impedance.

C. Radio Frequency (RF) Detection

Radio frequency (RF) is the oscillation rate of an alternating electric current or voltage or of a magnetic, electric or electromagnetic field or mechanical system in the frequency range from around 30 kHz to around 300 MHz

Radio Frequency Scanning (RFS) is a non-intrusive, in-service technique used to detect electromagnetic emissions from partial discharges (PD). By using wideband sensors on energized HV equipment such as GIS, AIS, transformers, and cables, RFS can quickly identify and localize PD sources.

RFS measures RF noise levels within defined frequency bands (typically 20 MHz–1000 MHz), enabling detection of medium to low-level PD activity. Establishing baseline noise levels improves immunity to external interference and supports risk assessment based on PD magnitude and repetition rate. The method is fast, cost-effective, and suitable for surveying large areas.

The **PDS100 RF Surveyor** detects, quantifies, and records RF signals, converting them into audio and numerical values for analysis and trending. Used with sensors such as UHF couplers, HFCTs, or antennas, it enhances PD localization—particularly in transformers and GIS. The UHF method is highly effective for locating internal insulation defects under live operating conditions and distinguishing PD from external noise through spectral analysis.

The system may also integrate acoustic emission sensing and live contact voltage measurement, providing a comprehensive approach to PD detection, localization, and condition assessment.

RF Signal Measuring Instrument

Doble Make Pds100/Pds200/Dfa300

(Partial Discharge Surveyor)

Purpose

- Used for non-contact detection and localization of partial discharge (PD) in high-voltage equipment such as switchgear, transformers, and cable terminations.
- Enables early identification of insulation degradation, tracking, and arcing to prevent catastrophic equipment failure.

Principle of Operation

- The DOBLE PDS 300 utilizes dual-sensor technology, incorporating both transient earth voltage (TEV) sensors and ultrasonic sensors (typically 40 kHz).
- The device captures electromagnetic and acoustic emissions generated by PD activity, converting these signals into visual and audible indicators that allow technicians to pinpoint the source of insulation stress.

Key Specifications

PARAMETER	SPECIFICATION
Detection Methods	Transient Earth Voltage (TEV), Ultrasonic, radio Frequency interference
Display	High-resolution colour LCD screen
TEV Measurement Range	0 to 60 dBmV
Ultrasonic Measurement Range	-6 to 70 dBµV
Frequency Response (Ultrasonic)	40 kHz (centre frequency)
Audio Output	Internal speaker or 3.5 mm headphone jack
Battery	Rechargeable Li-ion, up to 8 hours continuous use
Storage	Internal memory for data logging and screenshots
Communication	USB for data transfer and charging
Operating Temperature	-10 °C to +50 °C
Weight	Approx. 450 g

Operation Steps (SOP)

1. Press and hold the Power button to initialize the scanner.
2. Press Start/Stop button to start or stop the scan.
3. Perform a background noise measurement in an area away from the target equipment to establish a baseline.
4. For TEV detection, place the sensor head firmly against the metal cladding of the switchgear or enclosure.
5. Monitor the decibel (dB) levels on the display; significant increases over baseline indicate potential PD.
6. Use the Navigation buttons to toggle between TEV and Ultrasonic modes or to access the measurement log.
7. Capture and save readings for trending and reporting via the internal “Save” function.

8. Mode Change: Press the Mode button >Select F5 to switch modes (RF &Acoustic)> Select desired view mode using F1, F2, F3 and F4.
9. Audio Volume Adjustment: Use the Up/Down navigation buttons while in live measurement mode to adjust speaker/headphone volume.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power	Charge the battery via included power adapter or check if the power button was held long enough.
Excessive Background Noise	Ensure the sensor is not near Battery Charger or use the "Subtract Noise" function.
Data Not Saving	Check internal memory capacity or perform a system reset.
Weak Signal	Check sensor contacts (for TEV) or ensure line-of-sight to the source (for Ultrasonic).
Interference	Move away from fluorescent lighting or mobile devices that may cause electromagnetic interference.

Hotspot Monitoring Measuring Instruments

It is advantageous over traditional visual inspection and other tools because infrared technology does not have to be in contact with the component being monitored. IR thermography is a form of nondestructive testing that measures temperature variations of components and helps in thermal health level monitoring of the whole component. IR thermography is generally known in fields such as fire, thermal testing, thermal imaging and thermometry.

- It detects infrared emitted from an object.
- It measures the amount of infrared energy emitted.
- It displays as infrared images.

Thermal radiation is directly related to changes in the temperature of objects.

When, say, a component changes temperature, the amount of infrared radiation emitted by the component also changes.

This change in infrared energy is captured by the IR sensor which changes the energy into an electronic signal.

The electronic signal is then digitized and processed to produce a thermal image or thermogram.

In a thermogram, different colors represent different temperature levels and hot spots are clearly visible.

Abnormal heating patterns reveal overloaded or loose connections, unbalanced loads, failed components, or other problems.

Captured images and temperature data may then be analysed to obtain a temperature reading.

By comparing relative temperature levels and variations of a component to a known good reference image, conclusions can be drawn for that IR thermography.

Through proper interpretation, IR thermography helps detect hot spots, abnormal heating levels at electrical/mechanical equipment, defective loose connections, and hence the subsequent temperature difference between components.

Instruments Used to Capture Thermograms:

FLIR E75/E76: High-resolution tools for upstream network inspection and monitoring critical assets. They feature laser-assisted focus for precise hot spot identification.

Range: -20 °C to 650 °C

FLUKE Ti27: A ruggedized camera for harsh industrial environments and process control. Used primarily for diagnosing electrical assets.

- Range: -20 °C to 600 °C

STANLEY STAN ONE S280: A compact, focus-free imager for quick-scan troubleshooting. Ideal for identifying thermal anomalies in HVAC systems.

- Range: -20 °C to 550 °C

FLIR TG267: A versatile spot temperature tool used for facility maintenance. Efficient at identifying suspect electrical connections.

- Range: -25 °C to 380 °C

Visual Introduction to Instruments:

FLIR E-76

FLUKE Ti 27

FLIR E-75

STANLEY STAN ONE S 280

FLIR TG 267

Stanley S280 Pro – Handheld Thermal Imaging Camera

Manufacturer: Stanley Instruments, India

Type: Thermal Imaging Camera

Purpose

Used for non-contact temperature measurement and thermal imaging in electrical, mechanical, building, and maintenance inspections. Enables detection of hotspots, insulation defects, and preventive maintenance analysis.

Principle of Operation

The Stanley S280 Pro uses a 256×192 infrared focal plane array microbolometer sensor that detects thermal radiation in the 8–14 µm range. The sensor converts this infrared energy into electronic signals to create detailed thermal images displaying temperature variations, allowing users to visualize heat distribution and detect anomalies such as overheating or poor insulation.

Key Specifications

PARAMETER	SPECIFICATION
Infrared Resolution	256 × 192 pixels
Temperature Range	-20 °C to +550 °C
Accuracy	±2°C or ±2% of reading
Thermal Sensitivity (NETD)	< 40 mK @ 30 °C
Spectral Range	8 – 14 µm
Storage	8 GB Micro SD Card
Operating Temperature	-10°C to +50 °C

Operation Steps (SOP)

1. Press and hold the power button for 5 seconds to switch on the device.
2. Point the imager at the object and pull the trigger to capture a thermal image.
3. Adjust temperature range if readings exceed 160 °C using the Settings menu.
4. Use the navigation buttons to zoom, preview images, or adjust brightness.
5. Use the torch button to activate illumination in dark environments.
6. Select colour palette (Iron, Rainbow, Gray, etc.) for optimal visualization.
7. Save images in the device for analysis.

Common Errors/Troubleshooting Tips

ISSUE	REMEDY
No Power	Check battery charge or power button press duration (5 seconds).
Incorrect Temperature Readings	Verify emissivity settings and adjust temperature range.
Image Too Dark/Bright	Adjust display brightness from Settings > System Settings.
False Hotspots	Avoid reflections from shiny surfaces or use matte tape.
Image Not Saved	Check SD card space or format card if corrupted.

Basic Troubleshooting

Colour Palette Change: Press Select Button > Choose Palate from bottom Menu > Select Iron

Temperature Unit Change: Press Select Button > Choose Settings from bottom Menu > Select Unit > Select °C

Display Brightness Adjustment: Press Select Button > Choose Settings from bottom Menu > Select System Settings > Select Display Brightness > Select High/Medium/Low (Adjust based on Requirement).

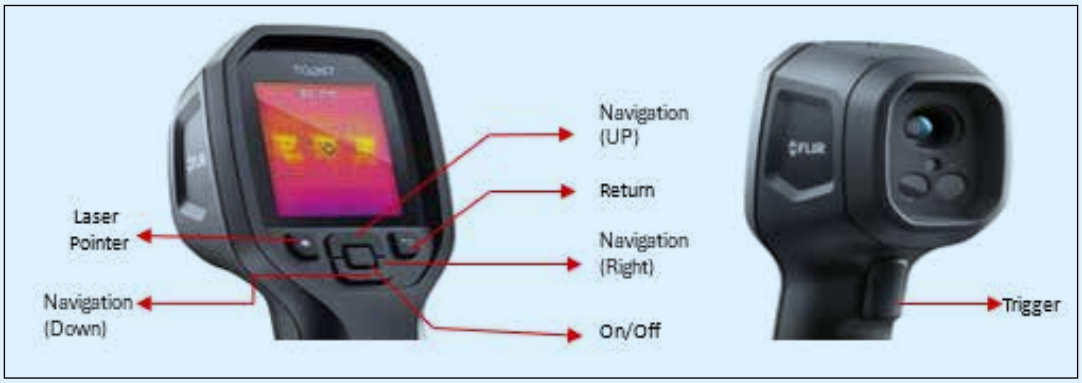
Temperature Range Selection: Press Select Button > Choose Settings from bottom Menu > Select Temperature scale > Select Temperature Range -20~150°C / 100~550°C/Auto (based on requirement)

Enabling High/Low Cursor Selection: Press Select Button > Choose Measurement > Enable Hot Spot / Cold Spot

Viewing Mode: Press Select Button > Choose View Mode > Select IR

FLIR TG267 – Handheld Thermal Imaging Camera

Manufacturer: FLIR Systems, USA



Purpose

Used for non-contact temperature measurement and thermal imaging, ideal for commercial electrical, facility maintenance, and HVAC applications. Enables detection of hot and cold spots that indicate serious issues, reducing diagnostic time and simplifying maintenance reporting.

Principle of Operation

The FLIR TG267 uses a 160×120 infrared focal plane array (uncooled microbolometer) sensor that detects thermal radiation in the 7.5–14 µm range. The sensor converts this infrared energy into electronic signals. The camera also features a 2 MP visual camera and FLIR's patented MSX® (Multi-Spectral Dynamic Imaging) technology, which embosses visual details onto the thermal image for greater clarity and context.

Key Specifications

PARAMETER	SPECIFICATION
Infrared Resolution	160 × 120 pixels
Visual Camera	2 MP (1600 × 1200 pixels)
Temperature Range	-25 °C to +380 °C
Accuracy	±1.5 °C (for 50°C to 100 °C) or up to ±3°C
Thermal Sensitivity (NETD)	< 70 mK
Spectral Range	7.5 – 14 µm
Storage	4 GB

Operation Steps (SOP)

1. Press and hold the power button to switch ON the device.
2. Point the imager at the object. The bullseye laser pointer indicates the centre of the measurement area.
3. Pull the trigger to capture and save a thermal image to the internal memory.
4. Use the navigation buttons to access the main menu to change settings or view the image gallery.
5. A short press of the "Back" button (which also serves as the power button) toggles the LED work light and laser pointer.
6. From the menu, select the desired colour palette (e.g., Iron, Rainbow, Grayscale) for optimal visualization.

Common Errors / Troubleshooting Tips

NO POWER	CHECK BATTERY CHARGE. CHARGE THE DEVICE VIA THE USB-C PORT.
Incorrect Temperature Reading	Verify emissivity settings are correct for the material. Ensure you are within the correct measurement distance (min 0.3m).
Blurry/Misaligned Image	Clean the IR and visual lenses. If visual details (MSX) are misaligned, adjust the focus.
False Hotspots	Avoid reflections from shiny (low emissivity) surfaces. Adjust emissivity settings if possible.
Image Not Saved	Internal memory (4 GB) is full. Connect to a PC to transfer or delete old images.

Basic Troubleshooting

Viewing Mode Selection: Press Select Button > Image Adjustments> Image Modes > Select MSX (Thermal + Visual)

Colour Palette Change: Press Select Button > Image Adjustments>Colors> Select Iron

Temperature Unit Change: Press Select Button > Navigate to Settings (gear icon) > Select General Settings > Select Temperature Unit > Select °C

Display Brightness Adjustment: Press Select Button > Navigate to Settings (gear icon) > Device Settings>Select Screen brightness > Select High/Medium/Low (Adjust based on Requirement).

Enabling Laser/Work Light: Press the Laser button/Back button to toggle the Laser Pointer and LED Work Light.

Viewing Saved Images: Press Select Button > Navigate to the Gallery icon > Use navigation buttons to scroll through saved images.

FLIR E75/E76 – Handheld Thermal Imaging Camera

Manufacturer: FLIR Systems, USA

Purpose

Used for advanced non-contact temperature measurement and diagnostics in electrical, mechanical, and building maintenance. Enables detection of hotspots, faults, and moisture intrusion. Designed for inspections from safe distances, ranging from roof checks to electrical components.

Principle of Operation

The FLIR E76 utilizes a **320 × 240 infrared focal plane array** (uncooled microbolometer) sensor to detect thermal radiation in the 7.5–14 µm range, converting it into electronic signals for thermal imaging. It features **MSX (Multi-Spectral Dynamic Imaging)** technology, which embosses visual details from the built-in 5 MP camera onto the thermal image for enhanced detail and perspective.

Key Specifications

PARAMETER	SPECIFICATION
Infrared Resolution	320 × 240 pixels
Display	4-inch, 640 × 480 LCD
Temperature Range	-20 °C to 120 °C & 0 °C to 650 °C
Accuracy	±2°C or ±2% of reading
Thermal Sensitivity	<40 mK @ 30°C
Spectral Range	7.5 – 14 µm
Field of View (FOV)	24° × 18° (Standard lens)
Battery	Li-ion, > 2.5 hours continuous use
Op. Temperature	-15 °C to +50°C

Operation Steps (SOP)

1. Press and hold the **Power** button to switch ON.
2. Point at the object. Use the **Laser** button for aiming and laser-assisted autofocus.
3. Adjust focus via **Autofocus** or the **Manual Focus Ring**.
4. Pull the **Trigger** to capture/save the thermal/MSX image.
5. Use the **Navigation Pad** to adjust settings.
6. Select **Colour Palette** (Iron, Rainbow, etc.) for optimal visualization.
7. Transfer data to PC or Mobile via **USB-C** or **Wi-Fi**.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power	Check battery charge and proper insertion.
Incorrect Readings	Verify emissivity and reflected temp. Ensure target is in focus. Avoid shiny reflections.
Image Blurry	Adjust focus (Auto/Manual). Clean lens gently.
False Hotspots	Avoid reflections from low-emissivity surfaces. Change angle.
Image Not Saved	Check SD card space and insertion.

Basic Troubleshooting

Colour Palette Change: Tap Screen > Select Colour Icon > Select Iron, Rainbow, etc.

Temperature Unit Change: Tap Screen > Settings (gear) >Device Settings> Units > Temperature > Select °C.

Display Brightness: Tap Screen > Settings > Device Settings > Display Settings > Adjust.

Temperature Range: Settings > Camera temp range > Select 0 to 650 °C (or Auto).

Viewing Mode: Tap Screen > Image Mode Icon > Thermal

Fluke Ti27 – Industrial Thermal Imager

Manufacturer: Fluke Corporation, USA

Type: Industrial Thermal Imager

Purpose

Used for industrial and commercial troubleshooting in electrical installations, electro-mechanical equipment, and process instrumentation. Enables detection of high-resistance connections, bearing failures, and unbalanced loads to prevent expensive downtime.

Principle of Operation

The Fluke Ti27 uses a 240×180 infrared focal plane array (uncooled microbolometer) sensor that detects thermal radiation in the 7.5–14 µm range. The sensor converts this infrared energy into electronic signals to create detailed thermal images. It utilizes **IR-Fusion** technology to align and blend visual and thermal images, allowing users to easily identify the exact location of heat anomalies.

Key Specifications

PARAMETER	SPECIFICATION
Infrared Resolution	240 × 180 pixels
Temperature Range	-20 °C to +600 °C
Thermal Sensitivity (NETD)	≤ 0.05 °C (50 mK) @ 30 °C
Spectral Range	7.5 – 14 µm
Field of View (FOV)	23° × 17°
Focus Mode	Manual Focus (Focus ring around lens)
Operating Temperature	-10 °C to +50 °C

Operation Steps (SOP)

1. Press the green power button to switch ON the device.
2. Remove the lens cap and point the imager at the target.
3. **Crucial Step:** Rotate the manual focus ring around the lens until the image is sharp and distinct.
4. Pull the trigger to freeze the image. Press **F1 (Store)** to save the image to the SD card.
5. Use the Up/Down arrow buttons to adjust the IR-Fusion blending (Visible vs Thermal) for better context.
6. Press the Menu button to select colour palette (Ironbow, Blue-Red, High Contrast, etc.).

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power	Check battery charge level on the battery's LED indicator. Ensure battery is clicked fully into place.
Blurry Image	The Ti27 is **Manual Focus**. You must rotate the lens ring; the camera does not autofocus.
Image Too Dark/Bright	Press "Auto" to reset level/span, or switch to Manual Level/Span to adjust contrast for specific targets.
Visual/IR Misaligned	Adjust IR-Fusion settings. At very close distances (<15cm), slight parallax error is normal.
Image Not Saved	Check if the SD card is locked (switch on side) or full.

Basic Troubleshooting

Colour Palette Change: Press Menu Button > Select Palette>Normal > Select Ironbow

Temperature Unit Change: Press Menu Button > Select Settings (F3) > Select Units(Auto/Manual) > Select °C

Display Brightness Adjustment: Press Menu Button > Background (F3) > Select Backlight > Select Up/Down.

Enabling High/Low Cursor Selection: Press Menu Button > Spot temp(F1) > Enable Centre Point, or Hot/Cold Markers (ON/OFF).

Ground Resistance Measuring Instruments

MECO 4680BL – Clamp-On Earth / Ground Resistance Tester

Manufacturer: Meco Instruments Pvt. Ltd., India

Device Interface

- **Power / ON-OFF:** Power button to switch the device on or off
- **HOLD:** Freezes the current reading on the display
- **FUNC / Mode:** Selects between resistance and current measurement modes
- **REC / MEM:** Used for data storing and recalling memory sets
- **ALARM:** Configures high/low alarm thresholds for resistance or current
- **Trigger / Jaw Lever:** Opens the 65 x 32 mm elliptical jaw to clamp around conductors
- **LCD Display:** 4-digit, 9999 counts backlit screen for viewing measurements

Purpose

- Used for non-contact ground resistance measurement of electrical installations like transmission poles, transformers, and lightning arrestors.
- Enables testing in multi-grounded systems without disconnecting the ground connection or using auxiliary rods.
- Detects leakage current (Model 4680BLC) and performs continuity loop tests for preventive maintenance.

Principle of Operation

- The MECO 4680BL operates by injecting a predefined high-frequency voltage into the ground circuit via a specialized clamp-on transformer.
- It then measures the induced magnetic current generated in the circuit to calculate resistance according to Ohm's Law (R = V/I).
- The system uses an amplifier with noise rejection to ensure accuracy even in environments with electrical interference.

Key Specifications

PARAMETER	SPECIFICATION
Ground Resistance Range	0.01 Ohm--1000 Ohm
Display	4-digit 9999 counts LCD with Backlight
Jaw Opening	Approx. 65 x 32 mm (Elliptical)
Accuracy	± (1% + 0.1Ω) for lower ranges; varies by range
Measurement Frequency	3.333 kHz
Memory	99 Sets of data storage
Conductor Size	Flat strips up to 65 mm; Round up to 25 mm dia.
Battery	1.5 V (AA) x 4 (Alkaline)
Operating Temperature	-10° C to +55°C
Weight	Approx. 1250 g (including batteries)

Operation Steps (SOP)

1. **Power On:** Press the power button. The device will perform a self-calibration check; wait for the "beep" indicating it is ready.
2. **Verify Accuracy:** Use the included Resistance Verification Plate (e.g., 5.1 Ω) to ensure the device is reading correctly before field use.
3. **Clamp Conductor:** Press the trigger to open the jaws and clamp around the grounding strip or wire. Ensure the jaws are fully closed.
4. **Read Measurement:** Observe the stable reading on the LCD. Use the HOLD button if you need to record the value in a difficult-to-see position.
5. **Data Storage:** Press the MEM/REC button to save the current reading to the internal 99-set memory.

Common Errors / Troubleshooting Tips

ISSUE	REMEDY
No Power / "B" Symbol	Check if batteries are inserted correctly; replace with four new 1.5 V AA alkaline batteries.
"OL" Symbol on Display	The measured value exceeds the range (overload). Ensure the jaw is properly closed around the conductor.
Inconsistent Readings	Clean the jaw mating surfaces with the provided cleaning brush to ensure a proper magnetic circuit.
False High Resistance	Check for noise signals or interference. Ensure you are not clamping onto a "dead" or non-loop line.
Calibration Error	Do not open the jaws during the initial self-calibration.

STANDARDS AND COMPLIANCE

Standards for Condition Monitoring Cell – CESC

Body Earth Resistance

The Condition Monitoring Cell adheres to **IS:3043 (1987)** standards for measuring equipment body earth resistance. In compliance with these regulations, all power equipment must be equipped with two distinct and separate earthing points. Their respective resistance values are detailed below:

Table-1: Equipment wise Body Earth Resistance

SL. NO.	ASSET NAME	RESISTANCE
1	Power Transformer (220/132/33 kV)	< 0.5 Ω (Both Side)
2	GIS (220/132/33 kV)	< 0.5 Ω (Both Side)
3	AIS (33 kV)	< 1 Ω (Both Side)
4	Power Transformer (33/11/6 kV)	< 1 Ω (Both Side)
5	33 kV RMU	< 1 Ω (Both Side)
6	AIS (11/6 kV)	< 1 Ω (Both Side)
7	Distribution Transformer	< 5 Ω (Both Side)
8	Consumer Switchgear	< 5 Ω (Both Side)
9	6-11 kV RMU	< 5 Ω (Both Side)
10	6-11kV Consumer Module	< 5 Ω (Both Side)
11	Pillar Box	< 5 Ω (Both Side)
12	PSS	< 5 Ω (Both Side)

Table-2: Location wise Body Earth Resistance

SL. NO.	ASSET NAME	RESISTANCE
1	132 kV and Above Sub-Station	< 0.5 Ω
2	33 kV Distribution Station	< 1 Ω
3	Downstream Assets	< 5 Ω

Partial Discharge Detection

We utilize five distinct methods to measure Partial Discharge in equipment through both direct and indirect monitoring. For all methods, a **Phase Resolved Partial Discharge (PRPD)** pattern is the primary indicator used to confirm the presence of a PD signal.

Ultrasound Detection

This technique is primarily used to inspect Air Insulated Equipment. It identifies PD by detecting distinctive "cracking" acoustic noises or specific PRPD patterns.

Table-3: Locations for ultrasound detection

SL. NO.	EQUIPMENT	LOCATION
1	Power Transformer	33/11/6 kV Air Insulated Cable boxes, Bushings, Cable Terminations
2	AIS Switchboard	Circuit Chamber, CB Chamber, Busbar Chamber, VT Chamber
3	33/11 kV RMU & Consumer Module	Circuit Chamber
4	33 kV GIS	VT Chamber
5	132/33 kV Dead Tank CT, VT	Transformer Tank (@25 kHz), Body
6	132/33/11 kV Capacitor Bank	bushings
7	6-11 kV Consumer Switch Gear	Circuit Chamber, CB Chamber, Busbar Chamber, VT Chamber
8	Distribution Transformer	HT Cable Boxes, HT FFA Chamber, Isolator Chamber, LT chamber (only for hybrid Transformer)
9	PSS	RMU cable box, PSS Body

Partial Discharge (PD) is identified by the distinctive 'cracking' noise on indicating instruments and specific **PRPD (Phase Resolved Partial Discharge)** patterns on locating instruments. Based on the signal intensity, findings are categorized as **RED, AMBER, or GREEN** according to the following criteria:

Table-4: Importance matrix of Ultrasound/ Acoustic Signal

SL. NO.	IMPORTANCE	ULTRASOUND INTENSITY
1	RED: Urgent Action Required	> 10 dB
2	AMBER: Close Monitoring Required	0-9 dB
3	GREEN: No action required	No noise / Mechanical noise

RFI Detection

This technique is specifically designed for inspections in high-noise environments, such as outdoor switchyards. It is used to monitor equipment including CTs, VTs, LAs, isolators, cable terminations, and circuit breakers, as well as 220 kV, 132 kV, and 33 kV bushings. The severity matrix is detailed below.

Table-5: Importance matrix of RFI Signal

SL. NO.	IMPORTANCE	OBSERVATION
1	RED: Urgent Action Required	20dB above base signal with distinctive time domain response
2	AMBER: Close Monitoring Required	Sub 20dB zone with distinctive time domain response
3	GREEN: No action required	Irregular time domain response

UHF Detection

This technique is applied to metal-clad equipment, including **220/132/33 kV Gas Insulated Switchgear (GIS)**, **33 kV Air Insulated Switchgear (AIS)**. A detected signal is validated as Partial Discharge (PD) only when a characteristic **PRPD (Phase Resolved Partial Discharge)** pattern is identified.

Table-6: Importance matrix of UHF Signal

SL. NO.	IMPORTANCE	OBSERVATION
1	RED: Urgent Action Required	50% and more signal with 900 MHz HPF 75% and more signal with 500 MHz HPF
2	AMBER: close Monitoring Required	Noise free signal 25%-50% irrespective of HPF
3	GREEN: No action required	Noise or no signal

TEV Detection

This technique is applicable to a broad range of equipment body and cabling systems. A signal is formally identified as Partial Discharge (PD) when a characteristic **PRPD (Phase Resolved Partial Discharge)** pattern is clearly observed.

Table-7: Importance matrix of TEV Signal

SL. NO.	IMPORTANCE	OBSERVATION
1	RED: Urgent Action Required	30 dBmV Above ambient
2	AMBER: close Monitoring Required	10-29 dBmV above ambient
3	GREEN: No action required	Below 10 dBmV signals

PD Detection using HFCT

This technique is specifically applied to cables featuring an **earthed sheath or screen**. A signal is officially classified as Partial Discharge (PD) only when a characteristic **PRPD (Phase Resolved Partial Discharge)** pattern is identified.

Table-8: Importance matrix of TEV Signal

SL. NO.	IMPORTANCE	OBSERVATION
1	RED: Urgent Action Required	30 dBmV
2	AMBER: close Monitoring Required	10-29 dBmV
3	GREEN: No action required	Below 10 dBmV signals

Infrared Thermography

We utilize Infrared Thermography to identify thermal anomalies across upstream and distribution assets. The following criteria define a RED (Urgent Action Required) observation based on temperature rise () or absolute temperature limits.

In Thermographic detection for,

1. Outdoor Upstream Assets

   Scope: 220/132/33/11/6 kV CSE, Bushing clamps, Pipe Bus joints, Copper Flexibles, CT/VT/LA terminals, Isolator Contact Jaws, Breaker Pads/Clamps, Sheath Earths (Indoor/Outdoor), Capacitor bank studs, and Bushing bodies.

   A RED condition is triggered if any of the following are met:

   a. Temperature Rise: > 40 °C above ambient temperature.

   b. Phase Differential: > 20 °C higher than corresponding phases.

   c. Absolute Temperature: 80 °C at any point.

2. For Power Transformers,

   Thermal monitoring for power transformers focuses on cooling efficiency and internal integrity:

   a. Cooling Circuit: If the temperature difference between the top and bottom header of radiator 10 °C indicate a choked cooling circuit.

   b. Internal Anomalies: A localized temperature rise of 20 °C on the Main Tank or OLTC (On-Load Tap Changer) is classified as an abnormal observation (RED).

2. For 220/132/33 kV GIS & RMU, any spot temperature or tank temperature difference >5 °C is considered as abnormal and marked as RED

3. For 33/11/6 kV AIS any localized hotspot temperature difference >10 °C is considered as abnormal and marked as RED.

4. For 11/6 kV consumer switchgear, RMU, Consumer Module localized temperature difference >10 °C is considered as abnormal and marked as RED

5. Distribution Transformers (11/6 kV)
   For distribution-level assets, the following thresholds indicate a RED status:

   a. Overall Body Temperature: Any reading > 60 °C.

   b. Main Tank Localized Hotspot: A temperature 15 °C higher than other areas of the tank body.

   c. Ancillary Chambers: Localized hotspots 15 °C higher than surrounding areas in:

      i. HT Cable Boxes

      ii. FFA Chambers

      iii. Isolator Chambers

   d. Pillar Box & LT Chambers: Any hotspot reaching 80 °C or higher.

> Try not to become a man of success, but rather try to become a man of value.
>
> — Albert Einstein

STANDARD OPERATING PROCEDURES

Standard Operating Procedure for Routine Inspection of Downstream Assets

1. **Purpose:** To outline the routine non-intrusive health monitoring (both status monitoring and condition monitoring) procedures for downstream assets in CESC distribution network covering distribution transformers (DTRs), packaged substations (PSS), ring main units (RMUs), consumer switch and consumer module at 11 kV and 6 kV level to ensure their reliable operation, detect faults early, and extend their operational lifespan.
2. **Scope:** This procedure is applicable to CM Cell & its vendor/ sub-vendor/ aggregator (business associates) developed by CM Cell.
3. **Responsibility:** The responsibility to follow this SOP shall rest with:

   CM Cell Officials, and

   Authorized Business Associates / Vendors engaged in inspection and audit activities.
4. **Procedures / Controls / Monitoring Process:**

ACTIVITY SEQ.	PROCESS DESCRIPTION
At Office (Before leaving for site inspection)	
1	Collect the list of assets to be inspected on a given date as per the given plan: **Indoor Teams:** based on Outdoor Job Diary **Outdoor Teams:** based on Monthly Asset Allocation list
2	Check the location of the assets to be inspected in CANS and finalize the route plan for asset inspection.
3	Check condition of the safety PPE, tools and equipment to be taken for site inspection as per the given plan in Annexure 6
3	Ensure availability of valid Authorization Letter from CESC along with company provided I-card and a Govt. I-Card. Carry out *Office TBM*, collect standard keys for RMU & MFPB, pack-up PPE (for every individual team member), tools and equipment; inspect vehicle condition, ensure availability of all statutory documents related to the vehicle and start for site inspection activities.

ACTIVITY SEQ.	PROCESS DESCRIPTION
At Site	
4	On reaching first site – post photographs of team members (wearing all necessary PPE) in the official designated Messaging group.
5	For every site – (after reaching asset location) check for the following: Check for unsafe conditions, hazards, non-standard lock and accessibility issue (if any). [***Note*** *– for NTR (No Test Return) cases - check for the relevant conditions as given in annexure 1 and post relevant image of the asset in the official designated Messaging group as NTR in consultation with CM Cell official.]* Check (a) whether any structure has got alive through modified tester (b) for any nearby live parts while reaching out to a pole mounted transformer through Live Line Detector. Carry out Site TBM and click image with the asset and then proceed for SM/ CM observations. For any unsafe condition or near miss at site report the same in the designated messaging group as per format in the *annexure 1.*
6	Carry out SM/ CM observation of the asset as per asset checklists and measurement guidelines (*annexure 1).*
7	For critical SM/ CM observations made at site post the same in the official designated Messaging group as per the Annexure 7 *[**Note:** Any other abnormal/ critical observation (not in the above list) made at site must be informed to CM Cell Official from site on priority basis.]*
8	After site inspection either carry out web-based data entry from site or carry out visit any of the CM Cell data entry center (list in *annexure 1*) and enter relevant observation in DREAMS portal within 3 working days along with relevant snaps of observations (for SM) and thermogram for hot-spot. For hot-spot observations greater than 150 deg C, send email to CM Cell along with relevant thermogram image and visual image. Note – Under SM observation it is mandatory to post thermogram of the conservator to capture oil level inside within the same.

Annexure 1

CM Cell Data Entry Center

STATION NAME	
1	Park Circus CM Cell Office
2	Kotrung Distribution Station
3	South Dum Dum Distribution Station
4	ESIC Joka S/S
5	Strand (North) Distribution Station

Format for reporting Near Miss/ Unsafe Condition

A. Near miss incidents format:

1. Name of the site.
2. Name of the victim.
3. Proper image.
4. A brief description of the incident.

B. Unsafe conditions format:

1. Name of the site.
2. Proper image.
3. A brief description of the unsafe condition.

Measurement Guidelines

MONITORING TECHNIQUE/ METHOD	THRESHOLD VALUE	STATUS TO BE TAGGED IN DREAMS
Partial Discharge – Ultrasound (Acoustic Mode at 40 kHz)	Case 1: with sound of crackling/ sizzling indicating electrical discharge)	
	PD Level > 10 dB	RED
	PD Level < 10 dB	AMBER
	Case 2: with mechanical sound (flat sound without variation)	
	Level > 20 dB	AMBER
Thermographic Scanning	Temp in DTR & MFPB > 70 deg C AND/ OR 40 deg C above ambient	RED

Annexure 2

Conditions for NTR (No Test Return)

CHECK POINTS	
1	Bare Wire Nearby
2	Blocked by Puja Pandal
3	Blocked by Stray Materials
4	Both Side Scaffolding Missing
5	Cat Ladder Blocked by Shop / Stray Materials

CHECK POINTS	
6	Cat Ladder Missing
7	General Cleaning Required
8	HT Side Inaccessible Due to Puja Hoarding
9	HT Side Scaffolding & Cat Ladder Missing
10	HT Side Scaffolding Damaged
11	HT Side Scaffolding Missing
12	LT Side Scaffolding Damaged
13	LT Side Scaffolding Missing
14	Non-Standard Lock-DTR
15	Non-Standard Lock-RMU/ Switch
16	O/T Door Jammed
17	On Roof Top
18	OT/T/H - HT Side Inaccessible Due to Space Constraint
19	OT/TH – Deweeding Required
20	P/T Covered in Vegetation
21	P/T HT Side Access Blocked
22	P/T Structure Tilted
23	RMU At O/T & T/H - Deweeding Required
24	RMU At P/T - Deweeding Required
25	RMU At Switch/Standalone - Deweeding Required
26	RMU Door Jammed
27	RMU Lock Jammed / Locking Arrangement Defective
28	Stray Materials In-Side Switch Room
29	Switch Shutter Jammed
30	T/H Shutter Jammed
31	Water Accumulation
32	X-Aligned

Annexure 3

SM/ CM Check Points for DTR

CM CHECK POINTS		
HT	1	Ultrasound Detection in HT Cable Boxes & Fuse Chamber
	2	TEV Detection in HT Cable Boxes & Fuse Chamber (if applicable)
	3	Measurement of Temperature at HT Chamber (Fuse Arrangement/ Cable Termination) (if applicable)
LT	1	Measurement of Temperature at LT Fuse Chamber
	2	Measurement of Temperature at Feeder Pillar Box
Earthing	1	Measurement of Body Earth Resistance
SM/ AO CHECK POINTS		
DTR	1	HT Cable Box (LHS & RHS) - Cable Gland Blank Off
	2	HT Cable Box (LHS & RHS) - Cover & Rain Guard condition
	3	HT Fuse Chamber - Condition of Cover, Explosion Vent, Rain Guard & T/L bolt
	4	Body - General Condition, Earthing.
	5	Sign writing - Body, HT Cable Box
	6	Cable Support Clamp - HT & LT
	7	For Oil Type - Oil Leakage, Condition of Breather, Guage Glass and Conservator Tank
MFPB/ PB	1	General Condition Body and Door Lock arrangement
	2	Cable Blank Off and Emergency Intake Blank Off
	3	Sign Writing and Earthing condition
	4	Unit Contact Condition
O/T Premises	1	General condition & access
	2	Compound wall condition
	3	Flooring
	4	Locking arrangement
	5	Premises name plate and danger plate
P/T	1	Access - Cat ladder
	2	Scaffolding
	3	Muffing
General		Any other abnormal observation

Annexure 4

SM/ CM Check Points for RMU/ Consumer Sw./ Consumer Module

CM CHECK POINTS	
1	Ultrasound Detection
2	TEV Detection
3	Measurement of Temp. of Different Chambers (if applicable)
	Measurement of Body Earth Resistance
1	Measurement of Temperature at LT Fuse Chamber
2	Measurement of Temperature at Feeder Pillar Box
SM/ AO CHECK POINTS	
1	Body - General Condition
2	Earthing - Condition
3	SF6 Gas Pressure Level
4	Circuit Cable Cover - Condition
5	Sign Writing including phase marking
6	Space Heater Status
7	Voltage/ Current Indication
8	FPI functioning
9	Door Lock arrangement
10	Cable trench condition
11	Accessibility; Civil condition of floor, bldg., wall.
12	Working platform
13	Any other abnormal observation

Annexure 5

SM/ CM Check Points for PSS

CM CHECK POINTS		
RMU Unit & Transformer Unit	1	Ultrasound Detection
	2	TEV Detection (if applicable)
	3	Measurement of Temperature (if applicable)
	4	Measurement of Body Earth Resistance
FPB	1	Measurement of Temperature

SM/ AO CHECK POINTS		
PSS Unit	1	Condition of PSS Body and its Plinth
	2	Door Arrangement
RMU Unit	1	SF6 Gas Pressure
	2	Earthing
	3	Physical Disposition of Feeders
	4	Cable Box Cover and Clamp
	5	Voltage Indication
	6	Sign Writing and Phase marking on cable box
	7	Cable Blank Off and Emergency Intake Blank Off
	8	Low FPI Battery Indication and Functioning of FPI Flags
	9	Isolator Status Indication
LV Compartment	1	MCB Labelling
	2	Supply Healthy Indication Lamp
	3	Voltage Indication
	4	Operating Handle
	5	Gland Plate Blank Off
Bldg. External	1	Condition of Wall, Door/ Shutter, Lock/ Latch, Danger Notice
Bldg. Internal	1	Condition of Wall, Ceiling, Floor, Lighting, Cable Trench and Inaccessibility
	2	Special events (if any)
General	1	Any other abnormal observation

Annexure 6

Safety PPE, tools and equipment/ instruments

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
1	Safety PPE as per approved PPE policy: Helmet, Shoe, Gloves, Eye Guard, Safety Belt, Ear Plug, Nose Mask	Condition – check for wear and tear (if any), best-before date (if any)
2	First Aid Box Items	Condition, Quantity of items (as per approved list) and Expiry date.
3	Live Line Detector & Modified Tester with crocodile clip	Condition and operation check

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
4	Infra-red Camera	Check (a) Power and Display (b) Image quality
5	Ultra-sound Measuring Device	Check power, screen display (if any) and audio
6	Earth Resistance Clamp Meter	Check power, display and operation with test resistance

Annexure 7

Critical SM/CM Observation

S. NO	CATEGORY	OBSERVATION LIMIT	NATURE OF POST
1	Partial Discharge in DTR, RMU, PSS (RMU Unit/ Transformer Unit), Cons. Switch, Consumer Module.	Mandatory for cases greater than 10 dB	Post visual image of the respective HT chamber with PD along with video file recording the PD sound
2	Any HT chambers – Door Open/ Any Live Parts of Equipment exposed to General public.	_	Post visual image of the respective HT chamber
3	Any live spark/ glowing live part or Temperature above 300 deg C or fire	_	Post relevant image (thermogram and/or visual, as applicable).
4	Asset severely damaged or tilted	_	Post relevant visual image

> Some people dream of success, while other people get up every morning and make it happen.
>
> — Wayne Huizenga

Standard Operating Procedure for PD Resolving & Re-Measurement (HS & PD) of Downstream Assets

1. **Purpose:** To outline the procedures for resolution/ re-measurement of partial discharge (PD) and/ or hot-spot (HS) in downstream assets in CESC distribution network covering distribution transformers (DTRs), packaged substations (PSS), ring main units (RMUs), consumer switch and consumer module at 11 kV and 6 kV level (wherever applicable).
2. **Scope:** This procedure is applicable to CM Cell & its vendor/ sub-vendor/ aggregator (business associates) developed by CM Cell.
3. **Responsibility:** The responsibility to follow this SOP shall rest with:

   CM Cell Officials, and

   Authorized Business Associates / Vendors engaged in inspection and audit activities.
4. **Procedures / Controls / Monitoring Process:**

ACTIVITY SEQ.	PROCESS DESCRIPTION
	At Office (Before leaving for site inspection)
1	a. Collect the list of assets (DTR/ RMU) from Mains-HT/ Mains-District/ S/S TG where PD resolution job will be taken up. Check for relevant historical data from DREAMS and keep note of the same for site reference. b. Collect the list of assets where re-measurement (of PD/HS)/ PD resolution is to be done on a given date. Make a list of the HT/LT chambers of the assets and its corresponding observations from DREAMS (historical data and work done details, if any) where re-measurement/ resolution is to be done.
2	Check the location of the assets to be inspected in CANS and finalize the route plan for asset inspection/ consult with Mains-HT/ Mains-District/ S/S TG for route finalization in case of resolution activities.
3	Check the condition of safety PPE along with relevant tools and equipment (as applicable for resolution or re-measurement), to be taken for site inspection as per the given plan. **(Annexure-1)**

ACTIVITY SEQ.	PROCESS DESCRIPTION
3	Ensure availability of valid Authorization Letter from CESC along with company provided I-card and a Govt. I-Card. Carry out *Office TBM*, collect standard keys for RMU & MFPB, pack-up PPE (for every individual team member), tools and equipment; inspect vehicle condition, ensure availability of all statutory documents related to the vehicle and start for site inspection activities.
AT SITE	
4	For every site – (after reaching asset location) check for the following: • Check for unsafe conditions, hazards, non-standard lock and accessibility issue (if any). • Check whether any structure has got alive through modified tester. • Carry out Site TBM. For any unsafe condition or near miss at site report the same in the designated messaging group as per format in the *annexure 1*.
5	**For PD Resolution Activities** **1. Pre-Shutdown PD Measurement** • Perform partial discharge (PD) measurements prior to the equipment shutdown. **2. During Shutdown** • After shutdown and opening of the HT compartment (in RMU/CM/Sw.Gr./DTR), ensure the compartment is externally short earthed. • Inspect for potential sources of PD. Look for: ▪ Moisture ingress/ condensed moisture, ▪ Surface tracking marks, ▪ Sulphation marks, ▪ Signs of insulation deterioration on cables, CTs (Current Transformers), VTs (Voltage Transformers), ▪ Cable crossover/ cable twist (if any), ▪ Touching cable boots inside RMU circuit chamber., etc. • Capture relevant images and share them in the designated messaging group. • During HV testing of CTs/VTs for condition assessment, maintain a safe distance from the equipment under test. Avoid standing directly in line of sight with the equipment—position yourself diagonally or to the side.

ACTIVITY SEQ.	PROCESS DESCRIPTION
	• Consult CM Cell official to determine suitable remedial measures, if required. • Ensure that the corrective actions taken at site are communicated to the CM Cell official. • Maintain a record of the remedial actions taken. **3. Post Shutdown** • After the HT compartment is boxed up and the system is re-energized, conduct PD measurement again. • Share the observations in the designated messaging group. **Note:** In case of any uncertainty or issues at the site, immediately consult a CM Cell official for guidance.
6	**For PD/ HS Re-measurement Activities** • Carry out relevant CM observations, take relevant thermogram and/or visual snap (for HS) and audio/ visual clip for PD. **Note:** For PD and critical HS (insulation burn/ glow) post the observation in the designated Messaging group.
7	**Data Entry in DREAMS** After re-measurement first allocate the asset for re-measurement entry and then enter data. Ensure the data is published within one working day. Post site resolving activity, ensure work done has been published by O&M and then proceed for asset allocation for re-measurement entry and then enter data. Ensure the data is published within one working day. **Note:** (a) For PD Resolution activities, post relevant site findings in the departmental database. (b) For Hots-spot re-measurement activities, prepare district-wise re-measurement report clearly indicating no. of asset inspected and assets with persisting observation along with relevant thermogram/ visual image as applicable.

Annexure – 1

Safety PPE, tools and equipment/ instruments

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
1	Safety PPE: Helmet, Shoe, Gloves, Eye Guard, Safety Belt	Condition – check for wear and tear (if any), Best Before date (if any)
2	First Aid Box Items	Quantity of items (as per approved list) and Expiry date
3	Live Line Detector & Modified Tester with crocodile clip	Condition and operation check
4	Infra-red Camera	Check (a) Power and Display (b) Image quality
5	Ultra-sound Measuring Device	Check power, screen display (if any) and audio

Standard Operating Procedure for Routine Inspection of Upstream Assets in Substations & Distribution Stations

1. **Purpose:** To outline routine, non-intrusive health monitoring procedures—including status and condition monitoring—for upstream assets within the CESC distribution network – SS, DS, RS and EHV Consumer. It encompasses 6, 11 & 33 kV Indoor Non-GIS Switchboards, 33 kV RMUs, 33, 132 & 220 kV GIS systems, Outdoor Yards (33, 132 & 220 kV), Power Transformers at DS/SS levels, and other associated HV/EHV equipment. The objective is to ensure reliable operation, enable early fault detection, and extend the operational lifespan of these critical assets.

2. **Scope:** This procedure is applicable to CM Cell & its vendor/ sub-vendor/ aggregator (business associates) developed by CM Cell.

3. **Responsibility:** The responsibility to follow this SOP shall rest with:

   CM Cell Officials, and

   Authorized Business Associates / Vendors engaged in inspection and audit activities.

4. **Procedures / Controls / Monitoring Process:**

ACTIVITY SEQ.	PROCESS DESCRIPTION
	At Office (Before leaving for site inspection)
1	Check allocated station name and team details from Outdoor job diary. Identify the list of assets to be inspected from DREAMS and collect instrument as per plan given in Annexure 1 - *Asset Category wise CM Instrument Matrix.*
2	Check condition of the safety PPE, tools and equipment/ instruments to be taken for site inspection as per the given plan in Annexure-5
3	Ensure availability of CESC I-Card/ valid Authorization Letter from CESC along with company provided I-card and a Govt. I-Card, whichever applicable. Carry out *Office TBM*, pack-up PPE (for every individual team member), tools and equipment; inspect vehicle condition, ensure availability of all statutory documents related to the vehicle and start for site inspection activities.

ACTIVITY SEQ.	PROCESS DESCRIPTION
	At Site
4	On reaching station – inform to Power Control Room before proceeding with inspection activities. Check for any unsafe condition/ any abnormality at site and carry out site TBM. For any unsafe condition or near miss at site, report the same in the designated messaging group as per format in the A*nnexure 3.*
6	Carry out SM/ CM observation of the asset as per asset-wise SM/ CM checklist and measurement guidelines (*Annexure 2*).
7	For critical SM/ CM observations made at site post the same in the official designated Messaging group as per the Annexure 6 **Note:** Any other abnormal or critically serious observations (not listed above) that impact equipment or human safety at the site must be promptly reported to the office.
8	After site inspection either carry out web-based data entry from site or enter relevant observation in DREAMS portal (from office) within 2 working days along with posting of relevant snaps of observations (for SM) & (a) Thermogram & Visual Image for hot-spot and (b) Phase Plot and image of relevant asset for Partial Discharge. Additionally for critical/ urgent SM observation and CM observations intimate respective O&M department through email within one working day from the date of inspection. **Note:** Critical SM Observation includes Improper Blank-Off of HT compartments, Bird Nest on live equipment, melting mastic in cable terminations, etc. CM Observation – Partial Discharge and Hotspot in HV/ EHV equipment, Radiator Bank Choked case for DS/SS level power transformers.

Annexure – 1

Asset Category wise CM Instrument Matrix

- For DS/ SS/ Sw. Stns. – All assets to be inspected.
- For EHT Consumer (Govt Body) – All CESC assets along with Consumer Sw. Boards (all panels) having CESC Supply as its Incomer.
- For EHT Consumer (Private Enterprise) - All CESC assets along with Incomer Sw. Panel *(ONLY)* of Consumer Switch Board.

ASSET CATEGORY	MONITORING TECHNIQUE/ METHOD	INSTRUMENT TYPE/ NAME/ MODEL	LOCATION OF MEASUREMENT
6/11 kV Indoor Switch Board	Partial Discharge - Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	CB chambers, Circuit Chambers and CT chambers (for separately mounted slip-on CT) of all panels along with Bus bar chambers (whichever accessible)
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	CB chambers, Circuit Chambers, LT Chambers and CT chambers (for separately mounted slip-on CT) of all panels along with Bus bar chambers including cable sheath connections to earth bar (wherever accessible)
	Body Earth Resistance	Earth Resistance Clamp Meter (ERCM) – MECO 4680BL	Panel earth bar at both end of the switchboard
	Transient Earth Voltage	UltraTEV Plus 2	Power cables & Panel Body (metallic enclosure) (wherever accessible)
33 kV AIS Switch Board	Partial Discharge - UHF	• PRPDMS - Phase Resolved Partial Discharge Monitoring System	CB chambers & Circuit Chambers of all panels (For detection - Hand-held sensor module to be placed on inspection window/ vent(louvres)/ bolted joints)
33 kV AIS Switch Board	Partial Discharge - Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	CB chambers, Circuit Chambers and CT chambers (for separately mounted slip-on CT) of all panels along with Bus bar chambers (whichever accessible)
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	CB chambers, Circuit Chambers, LT Chambers and CT chambers (for separately mounted slip-on CT) of all panels along with Bus bar chambers including cable sheath connections to earth bar (wherever accessible)
	Body Earth Resistance	Earth Resistance Clamp Meter (ERCM) – MECO4680BL	Panel earth bar at both end of the switchboard
	Transient Earth Voltage	UltraTEV Plus 2	Power cables & Panel Body (metallic enclosure) (wherever accessible)

ASSET CATEGORY	MONITORING TECHNIQUE/ METHOD	INSTRUMENT TYPE/ NAME/ MODEL	LOCATION OF MEASUREMENT
33 kV GIS/ RMU	Partial Discharge – RFI ***(Only for GIS)***	• DFA 300 • PDS 200 • PDS 100	All panels – front, rear and at sides (whichever accessible).
	Partial Discharge -Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	VT and CT chambers (for separately mounted VT/ slip-on CT outside panel enclosure) - for PD (whichever accessible)
	Gas Leakage detection	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	Gas Chambers for SF6 leakage detection when chamber gas pressure is found less.
33 kV GIS/ RMU	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	All chambers (cable chamber, CT/ VT chamber, isolator chamber, breaker chamber and bus bar chamber) including cable sheath connections to earth bar.
	Body Earth Resistance	Earth Resistance Clamp Meter (ERCM) – MECO 4680BL	Panel earth bar at both end of the switchboard
	Transient Earth Voltage	UltraTEV Plus 2	Power cables & Panel Body (metallic enclosure) (wherever accessible)
220/ 132 kV GIS	Partial Discharge - UHF	• PRPDMS - Phase Resolved Partial Discharge Monitoring System	For GIS mounted PD Sensors – PRPDMS to be directly connected; For Chambers without PD Sensors hand-held sensor module to be placed on metallic gas barriers for detection
	Gas Leakage detection	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	Gas Chambers for SF6 leakage detection when chamber gas pressure is found less
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	All chambers (cable chamber, CT/ VT chamber, isolator chamber, breaker chamber and bus bar chamber) including cable sheath connections to earth bar.
	Transient Earth Voltage	UltraTEV Plus 2	Power cables (wherever accessible)

ASSET CATEGORY	MONITORING TECHNIQUE/ METHOD	INSTRUMENT TYPE/ NAME/ MODEL	LOCATION OF MEASUREMENT
Outdoor Yard Equipment (220/ 132/33/11/6 kV)	Partial Discharge – RFI	• DFA 300 • PDS 200 • PDS 100	CB, Isolator, CT, VT, LA, Insulators, Cable Sealing Ends (CSE) (wherever accessible)
	Partial Discharge - Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic & Contact Mode) or • UltraTEV Plus 2 *(2nd Option*	Long Range Module (LRM) to be used for CSE (all voltage levels) & Supporting Insulators (at 33 kV & 6/11 kV level) (in Acoustic Mode) Remote Magnetic Transducer (RMT) to be used for dead tank CT/VT (at 132/220 kV Level) (in Contact Mode)
	Gas Leakage detection	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	SF6 filled CT when gas pressure is found less/ low.
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	All contact points in CB, Isolator, CT, VT, LA, Insulators, CSE / conductor joints
	Ultrasound Viewer (Partial Discharge – Image View)	FLIR PD Imager	CSE, CT, VT, CB, LA, Insulator, etc. (for confirmation of PD/ Gas Leakage (if any))
Capacitor Bank (220/ 132/33 kV)	Partial Discharge – RFI	• DFA 300 • PDS 200 • PDS 100	Supporting Insulators, NCT, Capacitor Units, Reactor
	Partial Discharge - Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	In Acoustic Mode -Long Range Module (LRM) to be used for Supporting Insulators, NCT, Capacitor Units, Reactor
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	All contact points/ joints in Capacitor Bank

ASSET CATEGORY	MONITORING TECHNIQUE/ METHOD	INSTRUMENT TYPE/ NAME/ MODEL	LOCATION OF MEASUREMENT
	Ultrasound Viewer (Partial Discharge – Image View)	FLIR PD Imager	Supporting Insulators, NCT, Capacitor Units, Reactor (For PD Confirmation Only)
Power Transformer & Earthing Transformer	Partial Discharge – RFI	• DFA 300 • PDS 200 • PDS 100	Oil – Filled Cable Box & OLTC
	Partial Discharge - Ultrasound	• UP9000 with ultrasound scanning module (in Acoustic Mode) or • UltraTEV Plus 2 *(2nd Option)*	Air – Filled Cable Box, Outdoor Bushing/ Outdoor Terminal Arrangement
	Ultrasound Viewer (Partial Discharge – Image View)	FLIR PD Imager	Outdoor Bushing/ Arrangement (For PD Confirmation)
	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera	Outdoor Bushing/ Arrangement, Cable Box, Radiator Bank, OLTC Chamber, Transformer Body
	Body Earth Resistance	Earth Resistance Clamp Meter (ERCM) – MECO4680BL	Transformer Body Earthing
	Transient Earth Voltage - Cables	UltraTEV Plus 2	Power cables (wherever accessible)

List of Abbreviations

1. PRPDMS - Phase Resolved Partial Discharge Monitoring System
2. PDS - Partial Discharge Surveyor
3. UHF – Ultra High Frequency
4. RFI – Radio Frequency Interference

Annexure – 2

Measurement Guidelines

MONITORING TECHNIQUE/ METHOD	THRESHOLD VALUE	STATUS TO BE TAGGED IN DREAMS
Partial Discharge - UHF	500 MHZ Band-Pass Filter – > 75 % 900 MHZ Band Pass Filter – > 50 %	RED
	No filter – 90 % 500 MHZ Band-Pass Filter – > 50 % 900 MHZ Band Pass Filter – < 25 %	AMBER
Partial Discharge - RFI	In Spectrum Analyzer Mode – > 20 dB above Ref. Line & in Time Resolved Mode – Irregular train of high magnitude pulses	RED
Partial Discharge – Ultrasound (Acoustic Mode at 40 kHz / Contact Mode at 25 kHz)	Case 1: with sound of crackling/ sizzling indicating electrical discharge)	
	• PD Level > 10 dB	RED
	• PD Level < 10 dB	AMBER
	Case 2: with mechanical sound (flat sound without variation)	
	• Level > 20 dB	AMBER
Ultrasound Viewer (Partial Discharge – Image View)	Heat Map Analysis in FLIR cloud server	For PD Location Confirmation ONLY
Thermographic Scanning	Temp – 40 deg C above ambient AND/ OR Temp difference between phases - > 20 deg C	RED

Annexure – 3

Data Capturing Format

Partial Discharge observation:

PLACE/ LOCATION OF DISCHARGE - (NAME OF THE EQUIPMENT WITH PANEL NAME & SWITCHBOARD NO./ FEEDER NAME WITH ODY NOMENCLATURE)	PHASE	VALUE IN DB	REMARKS

Hotspot Measurement:

PHASE	LOAD (A)	PLACE/ LOCATION OF HOTSPOT (PANEL/ FEEDER NAME)	TEMPERATURE (DEG C)	IMAGE NUMBER	REMARKS

Body Earth Resistance reading:

LOCATION/ ASSET DETAILS	BODY EARTH RESISTANCE (OHM)	REMARKS

Annexure – 4

Format for reporting Near Miss/ Unsafe Condition

A. Near miss incidents format:

1. Name of the site.
2. Name of the victim.
3. Proper image.
4. A brief description of the incident.

B. Unsafe conditions format:

1. Name of the site.
2. Proper image.
3. A brief description of the unsafe condition.

Annexure – 5

Safety PPE, tools and equipment/ instruments

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
1	Safety PPE as per approved PPE policy: Helmet, Shoe, Gloves, Eye Guard, Harness, Ear Plug (wherever applicable), Nose Mask	Condition – check for wear and tear (if any), best-before date (if any)

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
2	First Aid Box Items	Condition, Quantity of items (as per approved list) and Expiry date.
3	Partial Discharge Measuring Device	Check power, screen display and audio, whichever applicable. For DMS check power, sync status and signal generator power/ operation.
4	Infra-red Camera	Check (a) Power and Display (b) Image quality
5	Earth Resistance Clamp Meter	Check power, display and operation with test resistance

Annexure – 6

Critical Observation Reporting

S. NO	CATEGORY	NATURE OF POST
1	Any live spark/ glowing live part or Temperature above 300 deg C or fire	Post relevant image (thermogram and/or visual, as applicable).
2	Asset severely damaged or tilted or serious abnormality	Post relevant visual image

> Quality means doing it right when no one is looking.
>
> — Henry Ford

Standard Operating Procedure for Re-Measurement (PD/HS) & Resolution (PD) of Upstream Assets in Substations & Distribution Stations

1. **Purpose:** To outline procedures for resolution/ re-measurement of partial discharge (PD) and/ or hot-spot (HS) —for upstream assets within the CESC distribution network – SS, DS, RS and EHV Consumer. It encompasses 6, 11 & 33 kV Indoor Non-GIS Switchboards, 33 kV RMUs, 33, 132 & 220 kV GIS systems, Outdoor Yards (33, 132 & 220 kV), Power Transformers at DS/SS levels, and other associated HV/EHV equipment.

2. **Scope:** This procedure is applicable to CM Cell & its vendor/ sub-vendor/ aggregator (business associates) developed by CM Cell.

3. **Responsibility:** The responsibility to follow this SOP shall rest with: CM Cell Officials, and Authorized Business Associates / Vendors engaged in inspection and audit activities.

4. **Procedures / Controls / Monitoring Process:**

ACTIVITY SEQ.	PROCESS DESCRIPTION
At Office (Before leaving for site inspection)	
1	Collect the list of assets from Mains-HT/ S/S where PD resolution job will be taken up. Check for relevant historical data from DREAMS and keep note of the same for site reference. Collect the list of assets (from DREAMS) where re-measurement (of PD/HS) is to be done on a given date. Make a list of the assets and its corresponding observations from DREAMS (historical data and work done details, if any) where re-measurement is to be done. Check team details from Outdoor job diary and collect instrument as per plan given in *Annexure 1*.
2	Check condition of the safety PPE, tools and equipment/ instruments to be taken for site inspection as per the given plan in Annexure 4

ACTIVITY SEQ.	PROCESS DESCRIPTION
3	Ensure availability of CESC I-Card/ valid Authorization Letter from CESC along with company provided I-card and a Govt. I-Card, whichever applicable. Carry out *Office TBM*, pack-up PPE (for every individual team member), tools and equipment; inspect vehicle condition, ensure availability of all statutory documents related to the vehicle and start for site inspection activities.
At Site	
4	On reaching station – inform to Power Control Room before proceeding with inspection activities (in case of re-measurement activities). Check for any unsafe condition/ any abnormality at site and carry out site TBM. For any unsafe condition or near miss at site, report the same in the designated messaging group as per format in the A*nnexure 3*.
5	**For PD Resolution Activities** **1. Pre-Shutdown PD Measurement** • Perform partial discharge (PD) measurements of the plant prior to shut down. **2. During Shutdown** • After shutdown and opening of the HV chamber (wherever applicable), ensure the chamber/ plant is externally short earthed. • Inspect for potential sources of PD. Look for: ▪ Moisture ingress/ condensed moisture, ▪ Surface tracking marks, ▪ Sulphation marks, ▪ Signs of insulation deterioration on cables, CTs (Current Transformers), VTs (Voltage Transformers), etc. • Capture relevant images and share them in the designated messaging group. • During HV testing of CTs/VTs/Insulators/CB/ Panel for condition assessment, maintain a safe distance from the equipment under test. Avoid standing directly in line of sight with the equipment—position yourself diagonally or to the side. Use suitable equipment to pin-point the exact area of origin of PD to support O & M to execute the job. • Consult Office to determine suitable remedial measures, if required. • Suggest measures to O&M for PD resolution. • Ensure that the corrective actions taken at site are communicated to Office. • Maintain a record of the remedial actions taken. **2. Post Shutdown** • After the system is re-energized, conduct PD measurement again. • Share the observations in the designated messaging group. **Note:** In case of any uncertainty or issues at the site, immediately consult CM Cell official for guidance.

ACTIVITY SEQ.	PROCESS DESCRIPTION
6	**For PD/ HS Re-measurement Activities** • Carry out relevant CM observations, take relevant thermogram and/or visual snap (for HS) and audio/ visual clip for PD. **Note:** For PD and critical HS (insulation burn/ glow) observations - post the observation in the designated Messaging group.
7	**Data Entry in DREAMS** After re-measurement first allocate the asset for re-measurement entry and then enter data. Ensure the data is published within one working day. Post site resolving activity, ensure work done has been published by O&M and then proceed for asset allocation for re-measurement entry and then enter data. Ensure the data is published within one working day. **Note:** For PD resolution activities prepare a detailed report for departmental archiving. Refer *Annexure 2* for observation threshold values and DREAMS priority level.

Annexure – 1

Observation wise CM Instrument Matrix

For selection of instrument (category PD) – refer initial observation data.

OBSERVATION TYPE	MONITORING TECHNIQUE/ METHOD	INSTRUMENT TYPE/ NAME/ MODEL
Partial Discharge	Partial Discharge - UHF	PRPDMS - Phase Resolved Partial Discharge Monitoring System
	Partial Discharge – RFI	DFA 300/ PDS 200/ PDS 100
	Partial Discharge - Ultrasound (Acoustic Mode – 40 kHz Range)	UP9000 with ultrasound scanning module or LRM (Long Range Module)
	Ultrasound Viewer (Partial Discharge – Image View)	FLIR PD Imager (Note – To be used for PD source pin-point)
	Partial Discharge - Ultrasound (Contact Mode – 40 kHz Range)	UP9000 with Remote Magnetic Transducer.
	Transient Earth Voltage	UltraTEV Plus 2
Hotspot	Thermographic Scanning	Thermal Camera – FLIR/ Fluke Upstream Camera

Annexure – 2

Measurement Guidelines

MONITORING TECHNIQUE/ METHOD	THRESHOLD VALUE	STATUS TO BE TAGGED IN DREAMS
Partial Discharge - UHF	500 MHZ Band-Pass Filter – > 75 % 900 MHZ Band Pass Filter – > 50 %	RED
	No filter – 90 % 500 MHZ Band-Pass Filter – > 50 % 900 MHZ Band Pass Filter – < 25 %	AMBER
Partial Discharge - RFI	In Spectrum Analyzer Mode – > 20 dB above Ref. Line & in Time Resolved Mode – Irregular train of high magnitude pulses	RED
Partial Discharge – Ultrasound (Acoustic Mode at 40 kHz / Contact Mode at 25 kHz)	Case 1: with sound of crackling/ sizzling indicating electrical discharge)	
	• PD Level > 10 dB	RED
	• PD Level < 10 dB	AMBER
	Case 2: with mechanical sound (flat sound without variation)	
	• Level > 20 dB	AMBER
Ultrasound Viewer (Partial Discharge – Image View)	Heat Map Analysis in FLIR cloud server	For PD Location Confirmation ONLY
Thermographic Scanning	Temp – 40 deg C above ambient AND/ OR Temp difference between phases - > 20 deg C	RED

Annexure – 3

Format for reporting Near Miss/ Unsafe Condition

A. Near miss incidents format:

1. Name of the site.
2. Name of the victim.
3. Proper image.
4. A brief description of the incident.

B. Unsafe conditions format:

1. Name of the site.
2. Proper image.
3. A brief description of the unsafe condition.

Annexure – 4

Safety PPE, tools and equipment/ instruments

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
1	Safety PPE as per approved PPE policy: Helmet, Shoe, Gloves, Eye Guard, Harness, Ear Plug (wherever applicable), Nose Mask	Condition – check for wear and tear (if any), best-before date (if any)
2	First Aid Box Items	Condition, Quantity of items (as per approved list) and Expiry date.
3	Partial Discharge Measuring Device	Check power, screen display and audio, whichever applicable. For DMS check power, sync status and signal generator power/ operation.
4	Infra-red Camera	Check (a) Power and Display (b) Image quality

> An ounce of prevention is worth a pound of cure.
>
> — Benjamin Franklin

Standard Operating Procedure for Upstream & Downstream Job Quality Audit

1. **Purpose:** To outline procedures for upstream/ downstream job quality audit aimed at (a) assessing the quality and effectiveness of routine field inspections both upstream and downstream (b) identify gaps/ deviations and (c) promote continual improvement.

2. **Scope:** This procedure is applicable to CM Cell & its vendor/ sub-vendor/ aggregator (business associates) developed by CM Cell.

3. **Responsibility:** The responsibility to follow this SOP shall rest with:
   CM Cell Officials, and
   Authorized Business Associates / Vendors engaged in inspection and audit activities.

4. **Procedures / Controls / Monitoring Process:**

ACTIVITY SEQ.	PROCESS DESCRIPTION
At Office (Before leaving for site inspection)	
1	Identify assets (from DREAMS) inspected within last one and a half months against which no "work done" has been posted by concerned O&M team. Check team details from Outdoor job diary and collect instrument for routine SM/CM activity
2	Check condition of the safety PPE, tools and equipment/ instruments to be taken for site inspection as per the given plan: Annexure-3
3	Ensure availability of CESC I-Card/ valid Authorization Letter from CESC along with company provided I-card and a Govt. I-Card, whichever applicable. Carry out *Office TBM*, pack-up PPE (for every individual team member), tools and equipment; inspect vehicle condition, ensure availability of all statutory documents related to the vehicle and start for site inspection activities.

ACTIVITY SEQ.	PROCESS DESCRIPTION
At Site	
4	On reaching station check for any unsafe condition/ any abnormality at site and carry out site TBM. For any unsafe condition or near miss at site, report the same in the designated messaging group as per format in the A*nnexure 2*. **Note** For Upstream Station inspection inform Power Control Room prior to initiation of inspection
5	Carry out SM/ CM observation of the asset as per asset checklists and measurement guidelines. Refer Standard Operating Procedure for (a) Routine Inspection of Downstream Assets & (b) Routine Inspection of Upstream Assets in Substations & Distribution Stations for details.
6	**Report Preparation & Mismatch Analysis** Tabulate site inspection findings and compare them with observations recorded in DREAMS, using the format provided in the Annexure. Highlight any discrepancies between recorded and audited observations in red and count them accordingly. Calculate the total number of discrepancies and observations per asset to determine the mismatch percentage. Prepare vendor-wise asset inspection data and compute the overall mismatch percentage.

Annexure – 1

Job Quality Audit Format

SL. NO.	ASSET NO.	ASSET NAME	INITIAL INSPECTION DATE	INSP. BY	AUDIT DATE	SM OBS. FOUND BY VENDOR	SM OBS. FOUND DURING AUDIT	CM OBS. FOUND BY VENDOR	CM OBS. FOUND DURING AUDIT

CATEGORY	NO OF OBSERVATION AUDITED	NO OF DISCREPANCIES OBSERVED	%AGE DISCREPANCY
SM			
CM			

Annexure – 2

Format for reporting Near Miss/ Unsafe Condition

A. Near miss incidents format:

1. Name of the site.
2. Name of the victim.
3. Proper image.
4. A brief description of the incident.

B. Unsafe conditions format:

1. Name of the site.
2. Proper image.
3. A brief description of the unsafe condition.

Annexure – 3

Safety PPE, tools and equipment/ instruments

S. NO	PPES/ TOOLS/ EQUIPMENT	CHECKS TO BE PERFORMED
1	Safety PPE as per approved PPE policy: Helmet, Shoe, Gloves, Eye Guard, Safety Belt, Ear Plug, Nose Mask	Condition – check for wear and tear (if any), best-before date (if any)
2	First Aid Box Items	Condition, Quantity of items (as per approved list) and Expiry date.
3	Live Line Detector & Modified Tester with crocodile clip	Condition and operation check
4	Infra-red Camera	Check (a) Power and Display (b) Image quality
5	Partial Discharge Detection Device	Check power, screen display (if any) and audio
6	Earth Resistance Clamp Meter	Check power, display and operation with test resistance

ACKNOWLEDGEMENT

The preparation of this handbook has been a collective endeavour, and the Condition Monitoring Cell records its profound gratitude to the management of **CESC Ltd.** for their steadfast encouragement and unwavering support. Their foresight and commitment to technological advancement have provided the impetus for strengthening the framework of asset condition monitoring within the organisation.

We also extend our sincere appreciation to the various departments and field personnel whose practical wisdom, operational experience, and thoughtful insights have enriched this document. Their daily engagement with the complexities of power distribution and their dedication to maintaining the highest standards of reliability, safety, and operational excellence have been invaluable in shaping the procedures and practices presented herein.

The Condition Monitoring Cell further acknowledges the diligent efforts of its team members, whose perseverance in compiling, reviewing, and structuring technical knowledge has culminated in the creation of this handbook. Their commitment reflects a shared aspiration to institutionalise best practices and promote a culture of proactive asset health management.

It is hoped that this handbook will serve not merely as a procedural reference, but as a guiding compendium for engineers and technical professionals engaged in condition monitoring activities supporting the broader organisational pursuit of safe, reliable, and efficient operation of critical electrical infrastructure.

Condition Monitoring Cell
CESC Ltd.

Max [dB]: 32.0
Ultr
+
10.0
-
FLIR

Nothing is too wonderful to be true, if it be consistent with the laws of nature.

— Michael Faraday

If you cannot measure it, you cannot improve it.

— Lord Kelvin

Good fortune is what happens when opportunity meets with planning.

— Thomas Edison

CESC LTD.

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